



DELIVERABLE 2.6

Methodological Guidelines for the Identification and Involvement of DGs and Action Plans



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Executive Summary

This deliverable outlines the structurally Disadvantaged Groups (DGs) identified in other FITTER-EU tasks and deliverables so far, presents a literature review on the engagement of DGs, and provides the methodological guidelines for further mapping of DGs and for their engagement. In addition, this deliverable summarises the National Action Plans for the Identification and Involvement of DGs (NAP-DGs) developed in the frame of the six FITTER-EU country case studies (IE, DE, ES, IT, HU, PT). The analysis of DGs identified in the project so far shows consistency across different project initiatives in the groups identified in each of the four sectors of analysis (Energy, Transport, Building & Housing and Food & Agriculture), with a distinction between the *demand* or *consumer* side and the *supply* or *producer* side. The analysis also underlines the importance of an intersectional approach, as many DGs identified are characterised by an overlap of two or more dimensions of vulnerability. Based on aspects reported in the NAP-DGs and on a thorough literature review, this deliverable offers recommendations for the project's engagement activities in WP4, specifically in relation to the survey to be conducted in T4.1, and the FITTER Labs to be organised in T4.2. The proposed methodology hinges on interaction with intermediary organisations to ethically and successfully engage DGs in these activities.

Table of Contents

Executive Summary.....	3
List of Boxes	5
List of Figures	6
List of Tables	6
List of Acronyms.....	7
1 Introduction	9
1.1 Overview of FITTER-EU	9
1.2 Purpose and Scope of the Deliverable	9
1.3 Structure of the Deliverable	11
2 Who is the target population of the FITTER-EU project?	13
2.1 Preliminary identification of DGs in T2.1, T3.1 and national case studies	13
2.2 Cross-WP comparison of DGs identified	16
3 Literature review on the engagement vulnerable population groups	18
3.1.1 Introducing vulnerability	18
3.1.2 Challenges in engaging vulnerable groups.....	19
3.1.3 Principles for engaging with vulnerable groups.....	22
4 Analysis of the D2.6 NAP-DGs	29
4.1 Summary of DGs identified per sector and country	29
4.1.1 Energy.....	30
4.1.2 Transport.....	34
4.1.3 Building & Housing	35
4.1.4 Food & Agriculture	37
4.1.5 Cross-sectoral analysis	40
4.2 Intermediary organisations identified	48
4.2.1 Energy.....	48
4.2.2 Transport.....	50
4.2.3 Building & Housing	50
4.2.4 Food & Agriculture	51
4.2.5 Overview	52
4.3 Summary of national contexts for Stakeholder Engagement: Comparative analysis of challenges and incentives	59
4.3.1 Germany.....	59

4.3.2	Ireland	60
4.3.3	Spain	61
4.3.4	Italy	61
4.3.5	Hungary	62
4.3.6	Portugal	62
4.3.7	Common challenges	63
4.3.8	Common incentives	63
5	Guidelines for the engagement of structurally vulnerable groups in the FITTER-EU project.....	64
5.1	FITTER-EU strategy for the engagement of structurally vulnerable groups: the role of intermediary organisations.....	65
5.2	Recommendations to improve the mapping of intermediary organisations.....	68
5.2.1	Gaps between theoretical deliverables and the groups mapped in NAP-DGs	68
5.2.2	Gaps between Stakeholders Database and the groups mapped in NAPs.....	70
5.3	Guidelines for WP4 participatory activities.....	73
6	References	79
7	Annexes	83
7.1	Annex 1: National Action Plans for the Identification and Engagement of DGs	83
7.2	Annex 2: Preliminary identification of DGs	83
7.2.1	DGs identified in T2.1	83
7.2.2	DGs identified in T3.1	86
7.2.3	DGs identified in T3.2	91
7.2.4	DGs identified in ongoing sectoral case study work	92

List of Boxes

Box 1.	FITTER-EU project definition of “stakeholders”	13
Box 2.	DGs in Energy: Key action points for further mapping	68
Box 3.	DGs in Transport: Key action points for further mapping.....	69
Box 4.	DGs in Building & Housing: Key action points for further mapping.....	709
Box 5.	DGs in Food & Agriculture: Key action points for further mapping.....	70
Box 6.	DGs identified in the Stakeholders database: Key action points for further mapping	73
Box 7.	Key recommendations for stakeholder engagement of WP4 participatory activities.....	77

List of Figures

Figure 1. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per country.....	71
Figure 2. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per sector. *Some organisations are reported as relating to more than one sector.	72
Figure 3. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per DG	72

List of Tables

Table 1. List of FITTER-EU stakeholder engagement activities involving DGs	11
Table 2. Summary of potential DGs identified in different project tasks to date (only newly identified groups reported progressively).....	17
Table 3. Summary of DGs identified in the NAP-DGs in the Energy sector.	31
Table 4. Summary of DGs identified in the NAP-DGs in the Transport sector.....	34
Table 5. Summary of DGs identified in the NAP-DGs in the Building & Housing sector.....	36
Table 6. Summary of DGs identified in the NAP-DGs in the Food & Agriculture sector.....	38
Table 7. Cross-sectoral overview of DGs identified in the NAP-DGs in the Transport sector.	429
Table 8. Overview of intersectional DGs identified in the NAP-DGs.	473
Table 9. Overview of intermediary organisations identified in the NAP-DGs by national case study and DG.	58
Table 10. Challenges identified in the NAP-DGs by National Case Study.	63
Table 11. Incentives identified in the NAP-DGs by National Case Study.	63
Table 12. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per country and type of organisation	71
Table 13. Potential structurally disadvantaged groups in relation to the transition to electric mobility	94

List of Acronyms

AITEC	AITEC Asesores Internacionales SRL
CCGS	Cluster Community Group of Stakeholders
CSO	Civil society organisation
D	Deliverable
DE	Germany
DG	Disadvantaged Group
EP	Energy Poverty
ES	Spain
EU	European Union
EURECAT	Fundació Eurecat
FGB	Fondazione Giacomo Brodolini
GA	Grant Agreement
HU	Hungary
IDP	Internally Displaced People
IE	Ireland
IT	Italy
LGBT, LGBT+, LGBTQ+, LGBTQIA+	Lesbian, gay, bisexual, transgender, queer, intersex and asexual
NAP	National Action Plan
NAP-DG	National Action Plan for the Identification and Engagement of Disadvantaged Groups
PT	Portugal
SFC	Szekely Family & Co. Nonprofit Korlátolt Felelősségű Társaság
SGM	Sexual and Gender Minorities
SME	Small and medium enterprise
SV	Smart Venice SRL
T	Task
TFC	TFC Research and Innovation Limited
TUD	Technological University Dublin
UCM	Universidad Complutense de Madrid
UNINOVA	UNINOVA Instituto de desenvolvimento de novas tecnologias - Associação

WECF	Women Engage for a Common Future EV
WP	Work Package

1 Introduction

1.1 Overview of FITTER-EU

The FITTER-EU project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem (i.e., the FITTER ecosystem) that includes a digital platform to support policymakers in understanding which population groups may be at risk of being adversely affected by twin transition policies.

At the core of FITTER is the commitment to take an intersectional approach to tool-building that will assist policy development which addresses, or at least does not further exacerbate, societal inequalities across four sectors – energy, transport, building and housing, and food and agriculture.

The FITTER-EU project has six national case studies, each with one partner in charge of coordinating the related research activities. However, there are three countries where more than one partner contributes:

- Germany (WECF)
- Hungary (SFC)
- Italy (FGB), contribution of SV.
- Ireland (TU Dublin), contribution of TFC.
- Portugal (UNINOVA)
- Spain (UCM), contribution of AITEC and EURECAT.

There is also a sectoral division across national case studies, with each case study focusing on two sectors. All countries focus on Energy, while two countries focus on Transport (Spain - ES and Hungary - HU), two on Building & Housing (Germany - DE and Portugal - PT), and two on Food & Agriculture (Ireland - IE and Italy - IT).

1.2 Purpose and Scope of the Deliverable

In Work Package (WP) 2, *Conceptual framework and methodology*, the FITTER-EU project develops the conceptual and methodological framework that guides and informs the project's research. Task (T) 2.1, *Just transition definition and research design*, provided an operationalised definition of a 'just transition', while T2.2, *Governing the twin transitions*, developed a model of effective transition governance based on a mapping of structures and practices already in place in Europe.

T2.3, *FITTER expert group engagement strategy*, is contributing to WP2's aim of refining the research methodology and establishing a dedicated methodological approach for the engagement of stakeholders and targets from within the Cluster Community Group of Stakeholders (CCGS), as established in WP7, *Dissemination, communication and outreach*. The main output of T2.3 so far is Deliverable (D) 2.5, *FITTER Community Groups of Stakeholders Database, Manual and Action Plans*, which laid out three key components of the project's stakeholder engagement strategy:

1. The [Stakeholders Database](#), which details all stakeholders contacted and involved in the project and tracks their interest and participation in project activities.
2. The Manual, which covers the strategy for identifying and engaging with key stakeholders in the CCGS, including the Expert Group, to be consulted at several methodological steps throughout the project's implementation. The Manual comprises a methodological background, the FITTER-EU Principles for Stakeholder Engagement, the overall FITTER-EU Strategy and Engagement Plan, an overview of challenges and incentives, and engagement strategies per stakeholder type.
3. The National Action Plans, developed with each national case study leader, adapt the Manual to the particularities of each individual country (DE, IE, ES, IT, HU, PT), define a list of prioritised actors and an engagement action plan.

This deliverable is part of Task 2.4, *Engagement strategy for disadvantaged/vulnerable groups*, which includes the identification and involvement of the population targeted by the project, i.e., people negatively impacted by the policy measures established as part of the twin transition. This *at-risk* population constitutes the Disadvantaged Groups (DGs) that are prioritised by the project in the identification of hazards, negative impacts and risks, as well as the proposal of mitigation measures. These groups will be involved in the co-creation of the FITTER interventions through their participation in the FITTER Labs. While the Grant Agreement (GA) foresaw that the identification of DGs in T3.3 would be the main input for the start of this task, considering adjustments in the project timeline and the postponement of WP3, this deliverable has as main input preliminary findings of national case studies as part of ongoing work in WPs 2 and 3.

This deliverable aims to develop the methodology for the engagement of the target population of the project in each of the four sectors of analysis. The National Action Plans for the Identification and Engagement of Disadvantaged Groups (NAP-DGs) (Annex 1: National Action Plans for the Identification and Engagement of) were developed by partners as a tool to inform the preliminary identification of potentially vulnerable groups at the country and sectoral level. So far, the target population of the project has been referred to as “vulnerable groups” or “disadvantaged groups”, in line with the terminology used in the GA. However, based on the research conducted in Task 2.1—which has highlighted the systemic and structural nature of the conditions leading to vulnerability and disadvantage among populations affected by the twin transitions—these population groups will henceforth be referred to as structurally disadvantaged. That said, it is worth noting that in some parts of the project's delivery, the acronym DGs has been used, in order to align with the terminology adopted in other tasks within WPs 2 and 3 that are currently under implementation. In these cases, the acronym DGs should also be understood as referring to structurally disadvantaged groups.

This deliverable is strongly linked to WPs 3 and 4, as they include a considerable stakeholder engagement component with the project's target population (see activities listed in Table 1). The engagement activities foreseen with DGs include interviews, a survey and the FITTER Labs.

WP	Task	Activity according to GA	Revisions made during the project
WP3	T3.1	12 thematic focus groups (2 per country), 10 participants per group including civil society organisations (CSOs) representing vulnerable groups, UCM, M9-10	Replaced with online expert consultations via email.
	T3.1	60-semi structured interviews (10 in each country) with individuals belonging to DGs	Reallocated from T3.1 to T3.3
	T3.3	8 semi-structured interviews, EURECAT, M16-17	Unchanged
WP4	T4.1	Survey, SV and AITEC, M14-20	Unchanged
	T4.2	6 national FITTER labs (1 per country), each Lab consisting of 5 workshops, 15-20 participants per Lab, SV, M14-20	Delayed to M21-22
		1 online transnational FITTER lab, 30 participants from CCGS and DGs, SV, M14 to M20	Delayed to M23-24

Table 1. List of FITTER-EU stakeholder engagement activities involving DGs

1.3 Structure of the Deliverable

The deliverable is structured in four parts.

Chapter 2 presents a preliminary identification of the structurally disadvantaged cohorts targeted by the project –named DGs according to the terminology used in the GA–, based on work carried out in project activities so far (D2.1, D2.2, D3.1, T3.2 interviews and ongoing sectoral case study work). This chapter broadly shows consistency across different project activities in the potentially DGs identified, with a distinction on the *demand* or *consumer* side (i.e., end users of goods and services who may be at risk due to e.g. increased costs in the context of twin transition policies or facing exclusion by not being able to access sustainable modes of transport or energy) and those on the *supply* or *producer* side (i.e., workers in sectors and industries likely to be affected by twin transition policy and facing potential job loss). Chapter 2 also underlines the importance of an intersectional approach, as many instances of DGs identified are characterised by an overlap of two or more dimensions of vulnerability.

Chapter 3 reports findings from a literature review on the engagement of structurally disadvantaged groups in research activities, including an analysis of challenges and the proposal of principles for engagement.

Chapter 4 analyses the NAP-DGs specifically developed by partners as part of T2.4 to inform the preliminary identification of potentially vulnerable groups at the country and sectoral level. This chapter contains a summary of the DGs identified and the organisations mapped so far which represent their interests, along with summaries of partners' perceptions of possible challenges and incentives in the organisation of participatory activities with DGs (e.g., FITTER Labs).

Lastly, chapter 5 presents the proposed strategy for involving DGs in project participatory activities. This chapter, based on the analysis of the NAP-DGs (chapter 4) and the literature review (chapter 3) give some specific guidelines for further identification of relevant stakeholders representing structurally vulnerable groups, as well as give specific recommendations for the WP4 engagement activities –FITTER Labs and the T4.1 Survey.

2 Who is the target population of the FITTER-EU project?

Stakeholders of the FITTER-EU project include individuals and organisations who may be interested in the processes and outcome of the project, or who may be affected or perceive themselves as being affected by the project (D2.5). They include potential users of research findings or those who may benefit from them, as well as those who can support the development, implementation and evaluation of project activities. In the case of FITTER, stakeholders include several actor categories, including academia, industry, policy/government, social actors/community, and other European Union (EU) projects (D2.5). As outlined in D2.5, “structurally vulnerable groups affected by the twin transition” (p. 13, see Box 1) are one of FITTER’s key stakeholder groups. Previous project deliverables have carried out a preliminary identification of FITTER’s target population. This chapter summarises the groups identified so far.

Box 1. FITTER-EU project definition of “stakeholders”

The FITTER-EU project definition of “stakeholders” (using a broad definition) includes actors (individuals, groups or organisations) who influence and are influenced by the twin transitions, and those whose expertise (academic, practical or policy-focused) can help develop, implement and evaluate the FITTER-EU project, both as far as its research components are concerned and its innovations/IT based decision-making support system. In particular, the FITTER-EU project identifies the following stakeholder categories: civil society organisations, employer’s organisations, public service/policymaking, private sector, research/academia, trade unions and business angels. In FITTER-EU, stakeholders partly coincide with research subjects, i.e. structurally vulnerable groups affected by the twin transition.

This chapter presents a summary of the ongoing process for the preliminary identification of DGs, and presents a summary table of the groups identified so far.

2.1 Preliminary identification of DGs in T2.1, T3.1 and national case studies

Project tasks to date have included research towards the preliminary identification of structurally disadvantaged groups affected by the twin transition (See The template with the guidelines provided to partners for the NAPs-DGs can be found in the following link.

They are included in the project’s repository in the following path: “Documents/General/WP2 – Conceptual framework and methodology / T2.4 Engagement strategy for disadvantaged & vulnerable groups / National action plans for D2.6”

National action plans

- [Germany](#)
- [Hungary](#)
- [Ireland](#)
- [Italy](#)
- Portugal
- [Spain](#)

Annex 2: Preliminary identification of DGs).

T2.1 explored potential vulnerabilities uncovered and exacerbated by the twin transition and presented them in two deliverables: D2.1 and D2.2. While “EU equality law covers discrimination grounds based on sex/gender, race and ethnicity, age, sexual orientation, disability, religion or belief” (D2.1, p. 25), there are no universal definitions of these groups or agreed-upon criteria for identifying them. There is also considerable variability in understandings and approaches between countries. In FITTER, vulnerability is defined as “a multidimensional, differential, and dynamic concept, intimately linked with the persons’ individual characteristics and experiences in a certain environment (European Commission, 2021)” (D2.2, p. 9). Previous FITTER deliverables have problematised the concept of vulnerability and underlined a key issue:

[N]aming cohorts as vulnerable draws attention away from the systems and structures that produce advantage and disadvantage. It risks stigmatisation, ‘naturalising’ the vulnerability of these cohorts, and/or rendering them personally responsible for their adverse circumstances (D2.1, p. 76).

To account for this, FITTER has adopted the term *structurally vulnerable* or *structurally disadvantaged*. The structural inequalities identified in the project thus far are based on income inequality (and related aspects of poverty, socio-economic background and social exclusion), gender, age, migration background, disability, sexual orientation and gender identity, as well as geographical factors (e.g., regional differences, degree of urbanisation) (D2.1).

In the Annex, findings from D2.1 and D2.2 on identified vulnerable groups are reported per sector based on a literature review that included case study examples.

T3.1, and its findings as reported in D3.1, present the baseline: identified groups that are already disadvantaged in the context of each sector of analysis, and therefore their vulnerability in the context of twin transition policy should be researched, evaluated and understood. As a result, the focus is on existing vulnerabilities to energy poverty (EP), transport poverty, and disadvantage in relation to Building & Housing and Food & Agriculture, and whether these barriers can prevent specific groups from fully participating in the shift to more sustainable energy, transport, housing or food/farming, highlighting them as potential DGs in the twin transition context. Baseline disadvantage is also presented at the sectoral level.

In T3.2, 11 interviews with experts were carried out so far to better understand the twin transition and identify potentially disadvantaged groups in the context of the twin transition. The interview findings on the potentially disadvantaged groups identified are outlined in the Annex.

Ongoing sectoral case study work is proceeding to identify DGs in the context of the twin transition. Case studies are developed starting from policy objectives that were identified as common to the different countries in each case study (in T3.1). Following the choice of policy objectives, policy mechanisms were identified (T3.2 interviews). Then, case study reports for each sector examined potential DGs. The identified DGs in this task are closely linked to the objectives and mechanisms studied (they would be different if another objective or mechanism was used to determine the scope of the analysis).

Findings from T2.1, T3.1 and ongoing case study sectoral work are summarised in the following section.

2.2 Cross-WP comparison of DGs identified

The table below reports potential DGs identified in different project tasks to date (only newly identified groups reported for rows after T2.1). The different starting points and lenses adopted by each Task resulted in some additions being made progressively with each new task.

	Energy	Transport	Building & Housing	Food & Agriculture
T2.1	- Households with lower levels of education - Low-income households - Ethnic minorities, indigenous people and racialised people - Immigrants and those with low proficiency in the local language - Older people - People with a disability or chronic illness - Women - Young people and full-time students - Single-parent families - Socially isolated people and those living alone - Large and multi-occupancy households - People living in islands, small towns, suburban areas, disadvantaged regions - Families with an unemployed parent - Households using gas as fuel - Small and medium enterprises (SMEs) - People who need to spend long hours in the home	- Suburban and rural households - People with low income - People with a disability or chronic illness - Lone parent families - Older people - Women - People with ethnic minority background - People with low income and fossil fuel-run vehicles - Mechanics, logistics professionals, airport staff	- Tenants in the private market/social housing - Low-income households - Working class people - Low-income people - Racialised minorities - Children and youth - Older people - Women and gender-queer people - People with a disability or chronic illness - Migrant workers - Informal workers - Low-skilled workers - Homeowners - People living in inadequate housing	- Women in agriculture - Older farmers - Family-owned farms - Manual labourers - Rural communities - Small farms - Young farmers - Low-income consumers - Farmers in meat production

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	Energy	Transport	Building & Housing	Food & Agriculture
T3.1	• Unemployed people and those in precarious employment • People living in poorly insulated homes • Tenants in social housing and private market • Children • Rural households • Urban households	• Youth and students • Unemployed people • Informal workers • Those working non-standard hours • Non-native speakers	• Single-parent families • Victims of gender-based violence • Unemployed people • Immigrants • Refugees • Urban households • Rural households	
T3.2	• Low-skilled workers in extractive industries (coal, oil, and gas) • Workers in fossil-fuel-related infrastructure (boiler installers, gas station workers) • Highly specialised workers in outdated technologies (e.g., certain solar panel models) • People with low energy literacy	• Unskilled workers from automotive factories • Users that do not have or can't access charging points • Users without enough capital for the initial acquisition of an electric vehicle • Users without bureaucratic/grants knowledge • The cobalt value chain	• People who lack knowledge of grants and subsidies • Older people with low levels of education	• Manual crop workers, particularly seasonal and migrant workers in rural areas
Ongoing sectoral case study work	• Those living in multi-flat buildings		• Pensioners, students, trainees and the unemployed • People with low bureaucratic understanding and digital skills • Racialised people and people with a migration background • People living in urban areas • People experiencing homelessness	

Table 2. Summary of potential DGs identified in different project tasks to date (only newly identified groups reported progressively).

3 Literature review on the engagement vulnerable population groups

With the aim of maintaining an open approach in the literature review, the search for relevant bibliography focused on “engagement” and “vulnerability”. In particular, an exploratory literature review was conducted with Google Scholar, using the search string “how to engage” or “engaging”, and “vulnerable groups” or “disadvantaged groups”, and “policy” or “tools” or “design” or “environment” or “green transition” or “digital transition” or “environmental policy” or “digital policy” or “climate policy”. Beyond this, a separate search was carried out on the Taylor and Francis website in the CoDesign Journal (*CoDesign: International Journal of CoCreation in Design and the Arts*) with the keywords “vulnerable groups”, which yielded 75 results.

From these searches, 20 sources were selected, out of which 14 were journal articles (ranging from topics covering public services, (e-)health, education, engagement of the elderly, the (rural) youth, people with a disability, Base of the Pyramid individuals, and patients); 3 were policy papers (including a discussion paper and an analysis paper); 1 was a report; 1 was a book; and 1 was a book chapter. If search for definitions is included, there were 24 sources selected in total. Keywords: “vulnerable groups” and “definition” (2 books, 2 website entries, 1 journal article).

3.1.1 Introducing vulnerability

Vulnerability does not have a universally agreed-upon definition. It has been conceptualised in a variety of ways and in a variety of fields (e.g., research governance and ethical frameworks, health and social care discourses (Aldridge, 2015), humanitarian and legal perspectives), and its meaning varies according to the context it is used in (Larkin, 2009, p. 1, cited by Aldridge, 2015, p. 11). *Vulnerable groups* typically denote (Rogers, 1997, and Larkin, 2009, cited by Aldridge, 2015; Ippolito & Iglesias Sanchez, 2017; Rios, 2016):

- Racial and ethnic minorities.
- People with additional physical or cognitive needs.
- Persons at increased risk of illness.
- People on low income or who are unemployed.
- Persons who are incarcerated.
- People affected by unequal or dependent relationships;
- People living in sheltered accommodation.
- Migrants and internally displaced people (IDPs).
- Women.
- Sexual and gender minorities (SGM).
- Elderly individuals, youths and children.

Vulnerability is a consequence of social and systemic barriers such as illiteracy, language barriers, discrimination related to age, gender or ethnicity, and financial barriers caused, for instance, by low wages or insecure employment contracts (Moll et al., 2020). Moreover, everyone may be impacted by multiple vulnerabilities that may also be temporal (GOV.UK Office for Health Improvement & Disparities, 2022). For example, households living in poverty face daily living challenges, often have lower levels of education and are engaged in difficult, underpaid work. Many women carry a heavier burden in the household than men, ethnic minorities may be faced with segregated educational

environments, and older people living in low-income and/or rural, isolated areas are particularly vulnerable (European Commission, 2021).

Vulnerabilities often intersect and amplify each other's effects. For example, during a severe environmental event, the livelihood strategies of families in precarious economic situations often rely on women's domestic work and caregiving, further increasing women's (mental) health risks and exacerbating gender inequalities (Goudet et al., 2023, p. 16). Usually, the most vulnerable groups are the most excluded from policy processes because their needs can remain hidden as their representation and active participation in public and democratic decision-making is hindered by their difficulties (European Commission, 2021).

3.1.2 Challenges in engaging vulnerable groups

While there are multiple and competing definitions and considerations about what constitutes *vulnerable* or *disadvantaged* groups, the primary factor to keep in mind when implementing participatory activities or initiatives with them is not to put participants at further risk of harm and to ensure that vulnerability is not exacerbated by their participation. Furthermore, there needs to be a pronounced intention to “address inequality and powerlessness by giving 'voice' to individuals who may be marginalised or excluded” (Aldridge, 2015, p. 16), and ensuring “equality of voice” (Gherghina et al., 2021, p. 634).

Amann & Sleight (2021) categorise challenges in engaging with vulnerable groups in three types: resource-related challenges, initiation challenges and collaboration challenges. Based on the classification drawn up by Mulvale et al. (2019), two subcategories can be added to collaboration challenges: 1) (repeated) engagement challenges and 2) problems arising from power differentials.

Resource-related challenges could arise before, during, and after the activity, and these include limited funding, time, and personnel resources. Engaging with vulnerable groups can imply unexpected costs, e.g. travel or translation expenses (Amann & Sleight, 2021), and their effective engagement would often entail the presence of a dedicated research coordinator to reach out, provide information, support travel and build relationships (Mulvale et al., 2019a). Time constraints affect both participants and researchers (Forsythe et al., 2016). Vulnerable groups can often be considered “time poor” (OECD, 2024, p. 60), and participation might prove to be too resource-intensive for them. Moreover, they may have a more pronounced focus on day-to-day concerns and may not see the immediate value in participating (Mulvale et al., 2019a).

Challenges to initiating collaboration can be significant and these include problems associated with the identification and recruitment of the participants (Kiernan & McMahon, 2024) and issues of access to the target group due to a lack of contact information, inadequate infrastructure, consent challenges, etc. (Amann & Sleight, 2021).

Collaboration challenges can be organisational and can include difficulties such as defining suitable times and locations to meet contributors. A major collaboration issue refers to difficulties in attending meetings and participants dropping out – this phenomenon may have economic, social and/or physical circumstances (Amann & Sleight, 2021).

Participants may have a limited understanding of the project activities: this may be due to cultural, language or literacy barriers (Amann & Sleight, 2021).

Participants may not have the ability or the desire to express themselves. This could be due to several possible reasons:

- Limited understanding of the project aspects, unpreparedness or inexperience with co-production (Amann & Sleight, 2021).
- Lack of confidence in own abilities (Amann & Sleight, 2021), perception of low competence and expertise (Miscoiu & Gherghina, 2021).
- Issues with public speech and group situations (Miscoiu & Gherghina, 2021)
- Conflict-aversion (Miscoiu & Gherghina, 2021).
- Fear of stigma, harm, distrust in researchers' intentions, or (Amann & Sleight, 2021);
- Reluctance to challenge researchers due to inherent power structures (Amann & Sleight, 2021);
- Confusion or disinterest stemming from misalignment between their needs and the aims of the project (Jagtap, 2022).

Participants may feel overburdened by the collaborative process (Amann & Sleight, 2021), which can even lead to them dropping out (Mulvale et al., 2019b; OECD, 2024). Genuine inclusion can be challenging to achieve, as users can have complex needs and they might not have the capacity to participate fully in collaborative design due to reasons related to ability, access or socio-cultural factors (Hodson et al., 2023a). For example:

- Women often prove more difficult to involve due to gendered power relations, and higher time constraints due to care-work typically performed by them (Jagtap, 2022).
- Work and family commitments (Jagtap, 2022) clashing with private life (Miscoiu & Gherghina, 2021).
- Precarious employment situations (Mulvale et al., 2019a), e.g. irregular hours of work, or pressing need to find casual work (Jagtap, 2022).
- Health conditions (Kiernan & McMahon, 2024).
- Precarious housing situations (Mulvale et al., 2019).
- Location (Jagtap, 2022).
- Psychosocial deprivation (nervous and low in confidence, socio-cultural knowledge differences, constant stress and social isolation) (Jagtap, 2022).
- Changing circumstances (Kiernan & McMahon, 2024).
- Language barriers for migrant groups.

Group dynamics can take the form of stressors such as misunderstandings and frustrations, competing priorities, beliefs, values and attitudes, which can cause conflict (Amann & Sleight, 2021).

Participatory/deliberative events, while striving for emancipation, at the same time “(re)create domination” (Curato et al., 2019, pp. 1–2, cited by Mendonça & Asenbaum, 2025). This is especially the case if their design is not attentive enough to power differentials. It is important to note that

“requirements of rationality and impartiality” have consistently excluded marginalised groups (Mendonça & Asenbaum, 2025).

Power imbalances may prevent a certain level of trust necessary for effective participation to form (Forsythe et al., 2016) and could seriously hinder genuine inclusion and participation. These differentials refer to situations “when there are significant differences in educational and cultural backgrounds, language barriers, and gender imbalances” (Jiang et al, 2022, cited by Kiernan & McMahon, 2024). Participatory and co-design sessions are at high risk of domination by those of higher social standing, authority, with more knowledge about the topic, or of a certain gender (Gherghina et al., 2021; Jagtap, 2022), especially because active and meaningful participation might require cognitive, communication, conflict management and assertiveness skills that some groups may not have had the opportunity develop (Mulvale et al., 2019a). Power imbalances can arise both between the researchers/facilitators/organisers and the participants or between different participants or stakeholders (Kiernan & McMahon, 2024). Both the researcher's position and the participants' gender can affect the project processes and outcomes (Iivari et al., 2023). It is foreseen that in the Labs, specific attention would be needed to be paid to potential power imbalances between the participants representing those disadvantageously impacted by the twin transition, as it can possibly be expected that participants from very diverse socio-economic backgrounds will make up the sessions (e.g. one participant might be an elderly widowed woman from with an ethnic minority background, with low education living in a rural area, and another participant might be a middle-class cis, heterosexual, university-educated, male SME-owner from a metropolitan area). Power differentials might require specific measures when it comes to the interaction between the CCGS and the DGs as well.

Additional barriers to participation include the following:

- Digital technologies:
 - Older adults may face difficulties due to their capabilities (shaped by their age, technology literacy, access to information and communication technology), motivations (influenced by personal attitudes, education, income, social support), and factors such as accessibility and usability of the platform in question, access to a stable Internet connection, and connected expenses (e.g. devices, network subscription) (Backåberg et al., 2024);
 - Roma individuals (a 2020 survey by the EU Agency for Fundamental Rights found that only 43% of Roma households had Internet access and only 19% of Roma adults reported having basic digital skills. This disadvantageous position is exacerbated by technological, social, economic, institutional barriers disproportionately impacting Roma communities) (Jakupov, 2024).
- Inadequate accessibility, lack of tailored assistance (e.g. the location of the meeting does not have ramps).

Invitations may be rejected. This might be because of the conviction that involvement in the project would have a limited impact, due to a general rejection of political activities or owing to a feeling of powerlessness, and a fear that the sessions might be manipulated (Misoiu & Gherghina, 2021).

It has been shown that when participants feel like the project addresses their needs or issues, that they are properly included in the participatory process, and see the possible impact they might have on the process, they become very active. The activity may result in greater participation/engagement, which needs to be facilitated effectively (Jagtap, 2022)

Additional challenges may be in connection with ethical considerations, context and communication. (Mulvale et al., 2019a).

3.1.3 Principles for engaging with vulnerable groups

In general, it is necessary to focus on creating meaningful partnerships with participants instead of engaging them merely to fulfil a requirement (Forsythe et al., 2016). Three main principles of human-centred, participatory co-design can be listed (Steen, 2011, cited by Backåberg et al., 2024):

1. Understanding the needs, perspectives, lived experiences of people who will be impacted by the outputs designed;
2. Direct engagement to achieve this understanding;
3. Helping people express their needs, perspectives, and lived experiences through facilitation.

When it comes to vulnerable groups, key principles include:

- Patience, as working with DGs often takes more time than planned (Jagtap, 2022), “taking time to fully engage, listen for understanding and not move forward until participants or communities are ready” (Moll et al., 2020, p. 4);
- Balancing power structures, especially when participants are in completely different positions, and a *do-no-harm* approach: making sure that during the co-creation process, unequal structures are not strengthened (Iivari et al., 2023);
- Prioritising people over objectives or timelines (Mulvale et al., 2019a);
- Focusing on needs identified by DGs (instead of researcher- or system-identified ones) (Mulvale et al., 2019);
- Avoiding repeatedly engaging the same vulnerable participants (especially across long-duration interventions), which may lead to engagement fatigue (Kiernan & McMahon, 2024).

At the same time, it is necessary to strike a “balance between caring for vulnerable participants’ well-being and risking a ‘paternalistic mindset’ that prevents rather than supports capacity to participate” (Hodson et al., 2023a). Furthermore, the needs of those in power should not be neglected either (Iivari et al., 2023).

Before the session(s)

It is preferred to include vulnerable groups at important moments of the whole research process, as their absence at stages such as the definition of vulnerable groups can limit the later presence in co-creation (Hodson et al., 2023a). Previous research highlighted the need to “start early” and “ask often” when it comes to involving participants (Forsythe et al., 2016).

In the case of the most vulnerable participants, over-recruiting might be needed (Mulvale et al., 2019a).

It is recommended that training or orientation takes place as a preparatory step ahead of the session(s) (Mulvale et al., 2019a). This is necessary for both the participants and the facilitators.

In the case of the participants,

- Some authors suggest that pre-engagement activities should be organised in smaller, targeted groups to build relationships, establish trust, and manage expectations (Hodson et al., 2023a), using small-group activities (Dell et al., 2024a; Mulvale et al., 2019a).
- Training can be essential to address, on the one hand, the participants' lack of knowledge about both the co-creation process and the topics to be discussed (Jagtap, 2022), and on the other, their discomfort (Moll et al., 2020).
- Training programmes are recommended to be tailored to local conditions and characteristics of the targets, taking into account aspects such as educational level and age (Jagtap, 2022).
- Disseminating easy-to-access online information alongside the training is another measure that would enhance the inclusiveness of the process (Dell et al., 2024).

In the case of facilitators, it is necessary that they:

- Possess an in-depth understanding of the specific vulnerable groups (and the relevant cultural, socio-economic, health-related factors) (Amann & Sleight, 2021);
- Begin the process with “critical and embodied reflexivity”, that considers their own worldview, assumptions and values, their attitude and relationship to the participants and others involved in the process, and the system in which the participatory process will occur and critical awareness to outside pressures (Moll et al., 2020);
- Are committed to ongoing dialogue about various ways of knowing, the interrogation of power and privilege, and making the time and space for listening, sharing, critical thinking, learning and ensuring that diverse people are given an opportunity to participate (Moll et al., 2020);
- Are experienced in involvement methods, and are able to effectively mediate (Amann & Sleight, 2021);
- Are experienced in recognising power relations and equipped to handle them (e.g., “not allowing people to mansplain things or having more voice than others just because they think they could or making ironic or sarcastic comments” (Iivari et al., 2023, p. 312);
- Are experienced in co-creation as that leads to them being more equipped to identify obstacles in participation and in implementing strategies to overcome those obstacles (Jagtap, 2022);
- Are not alone in the process: multiple moderators are needed to balance biases (Dell et al., 2024)
- Have completed ethical training (Kiernan & McMahon, 2024).

It is also preferred that experts accompany facilitators to prepare for situations when moderators are unable to resolve misunderstandings (Dell et al., 2024).

As a preparatory step, it needs to be considered how someone e.g. with an impairment, a mental health issue, or someone who is homeless can take part in the research, and be prepared to put in place accessible methods and toolkits to enable participation (Kiernan & McMahon, 2024).

A decision needs to be made on the modality of the co-creation sessions, i.e., if they should be conducted online or offline. Usually, it is favoured that initial meetings take place in person, as face-to-face contact (especially) in the beginning can facilitate better collaboration (Forsythe et al., 2016), and it is deemed necessary for trust-building (Islind et al., 2023a). Online sessions have their advantages and disadvantages. Positive aspects include that they may be easier to attend for those living in remote areas, people with mobility issues or physical impairments that can be assisted via digital means, for people who would otherwise not have the time to attend an in-person event, or young people who are comfortable in digital spaces (Dell et al., 2024). Some evidence also suggests that people are more ready to express their views and engaging in political disagreements online (Dell et al., 2024).

Negative elements can include the fact that digital deliberation may be alienating for some as it requires digital literacy, equipment, and Internet connection. It should also be considered that even if a person has access to the internet—for instance, via a mobile device—this does not necessarily suit the design of a participatory online workshop, which may require the use of a computer. These barriers are especially prevalent in rural areas, among the elderly and/or people living in deprivation (Dell et al., 2024). There is some evidence of online participatory design settings proving to be problematic and challenging (Islind et al., 2023a).

When arranging the sessions, a core principle is staying flexible and adapting to the practical needs of participants. Significant attention should also be paid to designing the process to take place in a time-efficient way (Islind et al., 2023a), recognising the needs of the participants and developing strategies that maximise their input while minimising the time burden (Forsythe et al., 2016).

Scheduling meetings outside of traditional working hours (Forsythe et al., 2016) and providing flexible attendance options such as remote participation (Amann & Sleight, 2021), and alternating locations (Dell et al., 2024) and meeting times—morning, evening, lunch, weekend (Mulvale et al., 2019)—can serve well in enabling participants to attend the sessions. One element that might facilitate the organisation of a workshop or meeting is planning it back-to-back with other initiatives involving the participation of the DGs. For instance, if a representative organisation of DGs is already organising an event at a venue, the project's participatory activity could be scheduled in a dedicated timeslot within that event. For this a pre-requisite is the coordination and significative involvement with the DGs representative organisation, which might increase the probabilities of collaboration with the project.

For well-being purposes, it might be useful to ask participants to complete a “screener” survey to identify any issues or reservations (Hodson et al., 2023a). Interaction with gatekeepers (intermediary organisations who represent the interests of and/or work with specific DGs, supporting their involvement) when designing the sessions can be fundamental. Gatekeepers can help, for example, by identifying when it is/is not appropriate to involve specific participant cohorts as they have more complete understandings of participants' needs and capabilities, or, at later stages, can monitor participants' well-being to avoid engagement fatigue (Kiernan & McMahon, 2024).

During the session(s)

Overall, the literature consistently highlights “effort, attention, resources, and flexibility”, and sharing power as components required for successful engagement (Forsythe et al., 2016). These are necessary for what Kiernan & McMahon (2024) describe as *Development Ethics*: the aspiration to design inclusionary interventions that strive to ensure symmetrical relationships and equal power balances. For a “safe and inclusive space for collaboration”, sensitivity to power differentials, creative approaches might be needed (Moll et al., 2020).

Mutual understanding and managing expectations:

- it is necessary to establish solid relationships among the research team and the participants
- managing expectations: an important task already at the outset of the project is to provide information about the project and to manage expectations accordingly (Kiernan & McMahon, 2024); have a common understanding of what is possible, and acknowledge needs that cannot be met, and agree to central purpose guiding the project (Mulvale et al., 2019a); and establish commonly agreed-upon structures and terms of co-creation (Amann & Sleight, 2021)¹

In order to tackle power differentials between participants and the research team and to ensure meaning and purpose for participants (Mulvale et al., 2019a), it has been advised to establish some form of power sharing. Power sharing can take the form of formalised agreements for shared leadership, decision making and ownership of knowledge (Moll et al., 2020), open communication about documents, data and reporting, share leadership with a willingness to be challenged and directed, etc. (Mulvale et al., 2019a). Power sharing also has the benefit of enhanced participation as it facilitates participants to realise that their role is not symbolic but rather integral to the project’s development (Forsythe et al., 2016). Useful principles are listed below, grouped thematically.

Safety, openness, wellbeing

- Actively and respectfully listen to participants and make sure that staff does not take over (Mulvale et al., 2019a). Emphasis should be placed on fully understanding the lived experience and different motivations of DGs through conversation (Mulvale et al., 2019a), and identifying signs of discomfort (Hodson et al., 2023a).
- Need to be patient and wait for participants to be ready to share, and take time to “understand the person in their context” (Mulvale et al., 2019a).
- Understand and respect cultural differences (Mulvale et al., 2019a).
- Inform participants about possible negative after-effects of participation and provide mitigating advice (negative after-effects include feeling stressed and exhausted after the experience, or becoming resentful due to not having been taken seriously or given enough attention) (Dell et al., 2024).

¹ When it comes to managing expectations, it has been noted as a common problem to participatory processes that participants expect more influence than they will have (Brandsen et al., 2020, p. 7), which underlines the importance of communicating the exact roles.

- Legitimise the opinions of the participants by ensuring they are heard, and their opinions/perspectives/feedback considered as the project progresses (Kiernan & McMahon, 2024).
- Need to balance safety and openness: the opinions about calling participatory events “safe spaces” are mixed as it has been observed that it could alienate participants who felt that it censored or curbed their input (Dell et al., 2024). An alternative to this could be the establishment of a code of conduct that does not allow hate speech, and to have a facilitator that can be turned to in confidence if someone feels unsafe or unhappy with the rules of conduct (Dell et al., 2024).
- Make participants aware of moderator bias (Dell et al., 2024).
- To ensure the wellbeing of participants, it has been suggested to:
 - Encourage them to discuss involvement with a trusted individual (Amann & Sleigh, 2021).
 - Provide the presence of a counsellor there for anyone who might need help (Hodson et al., 2023a).
 - Provide emotional support and a quiet space for retreat (Mulvale et al., 2019a).
- Design informative materials to be accessible and inclusive (made in multiple languages, and/or with imagery of multicultural environments, including functionalities that help participants with physical impairments interact with them) (Dell et al., 2024).

Location

It is recommended to meet the participants where they are, in locations and environments where they are comfortable (Kiernan & McMahon, 2024). Holding the sessions in diverse locations can help remove barriers to participation. Dell et al., p. (2024b, p. 28) provide a useful summary about some measures that can be taken to maximise the accessibility of the events:

- “Choosing wheelchair-accessible spaces;
- Reserving first-row seats for people with special needs;
- Having publications in braille;
- Having websites and documents that use easily comprehensible language;
- Designing spaces to accommodate specific anxieties and sensory overload – think of poor acoustics or flickering lights”.

Tools

Tools identified in the literature underline the following aspects:

- Importance of using straightforward, clear, understandable, communal, plain language, to make it easier to search for, find, read, and understand information, ideally combined with visual elements (Kiernan & McMahon, 2024; OECD, 2024; Jagtap, 2022);
- Narrative modes of communication using concrete explanations and examples, without using abstract discussions (Jagtap, 2022);
- Other discursive strategies such as referencing local issues, using humour, etc. (Dell et al., 2024);

- Goal to create a relaxed, flexible and fluid environment (Smith et al., 2022);
- Adopting non-stigmatising options for data-sharing (Mulvale et al., 2019a), e.g. using personas instead of asking individuals to speak about personal experiences (Amann & Sleight, 2021) can help participants feel more at ease and enable them to be more active in the co-creation process (Hodson et al., 2023a);
- In the case of youth: need for fast-paced and short activities to maintain engagement and interest (Smith et al., 2022);
- Interactive, different types of tasks preferred “to engage all types of people” (Smith et al., 2022)
- Use of arts-based and creative tools (video, storyboards, clay, Lego (Moll et al., 2020), writing (Laursen, 2024), drawing, theatre, role-playing, music sessions (Dell et al., 2024) can be particularly effective to engage participants who may have difficulty with purely verbal interactions (Moll et al., 2020));
 - For icebreaking (e.g. drawing) (Dell et al., 2024);
 - To generate initial insights (Moll et al., 2020): observation that often creative tools (‘doing’ and ‘making’) can evoke deeper insights and creative solutions (Moll et al., 2020), and the ideas generated in such a way might transcend what people might put into words;
 - However, knowing the audience is essential: if methods are playful and open, “that may be too much for certain people, who are more used to talking and making their points” (Iivari et al., 2023, p. 312).
- Other nonverbal deliberation exercises could help with mitigating the risk of someone dominating the discussion (e.g. designating different corners of a room to different viewpoints and asking participants to stand near the perspective they agree with most (Dell et al., 2024);
- Online tools could help with:
 - Keeping track of speaking time, preventing any one participant from dominating the discussion (Dell et al., 2024);
 - Content moderation, by limiting hate speech, slurs and abusive language (Dell et al., 2024);
 - Summarising and visualising discussions (e.g. via argument maps), making the deliberation easier to understand (Mikhalovskaya, 2024; Landemore, 2022, cited by Dell et al., 2024);
 - Screening for and removing misinformation, detecting logical fallacies, and providing easy access to relevant information (Novelli & Sandri, 2024, cited by Dell et al., 2024).
- Voting - various findings:
 - Consensus decision-making protects numerical minorities (Mendelberg et al., 2014, cited by Dell et al., 2024), and adds weight to under-represented group members’ views (Dell et al., 2024);
 - Consensus decision-making tends to suppress minority voices (Mendonça & Asenbaum, 2025).

Continuous evaluation

In the example referred to by Backåberg et al. (2024), after each workshop, the facilitation team conducted a debriefing session to reflect both on the structure of the facilitation and the content of the discussions to gain insights into the participants' experiences. Such feedback collection exercises could take place both anonymously or face-to-face (Mulvale et al., 2019a). Dell et al. (2024) have suggested a similar session, but with the participation of all organisers and partnering organisations. In both examples, the inputs were used to make refinements to the workshops (e.g. re-evaluation of the activities to be included, how they were introduced).

Recognising contributions

Taking part in the project's participatory activities is not cost-free for participants and especially vulnerable groups could need additional support or tailored measures to be able to participate in an equal and effective way (OECD, 2024, p. 59). Incentives can enable continuous engagement, create a more inclusive process, and build trust. Support for participation can take many forms, including financial incentives (e.g. gift cards), offering food, transportation assistance (e.g. free bus passes, taxi vouchers, or reimbursing transportation costs), childcare, flexible scheduling to accommodate working hours, offering training or educational opportunities, providing information in multiple languages (European Commission, 2021; OECD, 2024, p. 59), providing the assistance of accompanying persons to support individuals with special needs, having on-site cultural mediators or translators, or collaborating with social assistance departments to provide psychosocial support to groups such as ex-prisoners or Roma communities (Dell et al., 2024).

Specifying shared roles and responsibilities, encouraging participants to recognise that they are making a difference (Mulvale et al., 2019a), and making them feel heard and taken seriously (Amann & Sleight, 2021) could serve both as an incentive and reward. It has been highlighted several times that when participants recognise their impact and when the project's activities matter to them or they are dependent on its outcomes, their engagement significantly increases (Jagtap, 2022).

Similarly, acknowledging the participants' inputs is of crucial importance. One way of achieving this is through continuous, two-way feedback loops, by providing follow-up updates (e.g. by collecting feedback from participants and using them for a report to be sent after the events (Dell et al., 2024); and especially on the final developed concept) (Kiernan & McMahon, 2024). Others suggest "establishing formal recognition for the value of service user input" (Moll et al., 2020). Additionally, participation can be rewarded materially (e.g. by gifting useful items to mark milestones in engagement (European Commission, 2021), or giving out, for example, thank-you certificates (Amann & Sleight, 2021)(Amman & Sleight, 2021).

Ethical considerations

It is necessary to think of consent as dynamic, acknowledging that it may change over the process, thus "ongoing informed consent and clear information loops throughout are essential, so a participant is empowered to continue or withdraw at any stage of the process" (HSE, 2023 cited by Kiernan & McMahon, 2024). It is necessary to review consent at various stages of the process, and ask for

renewed consent when parameters of the project, or the type and level of activities, change (HSE, 2023, cited by Kiernan & McMahon, 2024).

There is a recommendation for dynamic ethics approval processes where “protocols can be revised as engagement, data collection and feedback loops shift and change” (Kiernan & McMahon, 2024), and which are “reflexive and responsive to issues as they arise during the design research process” (Kiernan & McMahon, 2024).

After the session(s)

Suggested follow-up measures include (Dell et al., 2024):

- Archiving the process and lessons learnt;
- Making sure any dissemination activities reach the affected communities.

After a participatory session, it is crucial to account for the outcomes of the participatory action to the involved DGs. In particular, the results and outcomes of the initiative should be shared in accessible formats, allowing the possibility for feedback and input collection from the participants who took part in the process. Additionally, it is essential that the support and contribution of the actors participating in the co-creation activities are recognised and valued, promoting ownership and supporting the visibility of the vulnerable groups involved—if this is their willingness. Finally, it is important to co-create the next steps and be transparent about the expected future activities.

Some of the recommendations and challenges outlined in this chapter are revisited in chapter 5, which focuses on providing practical guidelines for the involvement of structurally vulnerable cohorts targeted by the FITTER-EU project. The following chapter, which analyses NAPs-DGs, serves as a fundamental input for the last chapter of D2.6, analysing the most relevant structurally vulnerable cohorts that has been identified so far in the six FITTER-EU project case studies.

4 Analysis of the D2.6 NAP-DGs

4.1 Summary of DGs identified per sector and country

In the NAP-DGs, country case study leaders were asked to provide a preliminary summary of the structurally disadvantaged groups identified in work conducted as part of previous and ongoing project tasks. The aim of this exercise was to prepare the ground for upcoming stakeholder engagement efforts involving identified groups (in particular, as part of WP3 and WP4 activities, encompassing interviews, a survey and the FITTER Labs).

Each country case study focuses on two sectors. All countries focus on Energy; in addition, Spain and Hungary focus on Transport, Germany and Portugal focus on Building and Housing, and Ireland and Italy focus on Food and Agriculture. The tables below summarise the DGs identified in the NAP-DGs, with a distinction between the *demand side* and *supply side* which has emerged in project conceptualisations so far. The *demand side* is understood as a lens to view the sector through the consumers’ perspective, e.g. users of energy or transport services. The *supply side*, instead, considers

possible adverse effects of the twin transition for workers in each sector (e.g., workers in the fossil fuel industry who may face job loss).

4.1.1 Energy

Summary table of DGs identified

Potentially disadvantaged groups		IE	DE	ES	IT	HU	PT
Demand side	Low-income households	X	X	X	X	X	X
	Older adults	X	X	X	X	X	X
	Young people			X			
	Women	X	X	X	X		
	Rural households			X	X	X	X
	People with migrant background	X			X		
	Roma people	X			X	X	X
	People with disabilities	X		X	X		
	Carers	X					
	People not in employment	X		X			
	People renting in the private market	X	X				
	People with low level of education	X					
	Single parents	X					
	People living in older buildings		X				
Supply side	Workers in the fossil fuel industry		X	X	X		X
	Workers in the peat industry	X					
	Workers with low digital literacy	X			X		

	Inhabitants of former fossil fuel regions		X				X
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Table 3. Summary of DGs identified in the NAP-DGs in the Energy sector.

Demand side

Low-income households and their intersections

Low-income households are the most affected by EP, as they allocate a higher share of their income to energy costs. The German NAP-DG reported that affordability is of key importance, as some “socio-economic groups such as low-income households and tenants are massively affected by EP” (p. 4).

The Italian NAP-DG, which highlighted EP as an ongoing issue, reported that “low-income groups are more likely to face difficulties in affording new technologies that are part of the digital energy transition, such as smart meters, home automation systems, or energy-efficient appliances” (p. 4). Similarly, the Hungarian NAP-DG reported that EP “is a major issue, with many low-income households unable to afford energy-efficient upgrades (e.g., insulation, modern heating systems)” (p. 3). In addition, “dependence on solid fuels (wood, coal) and outdated heating systems increases exposure to air pollution and health risks” (p. 4). The Portuguese NAP-DG reported that:

[T]he transition to greener energy sources may temporarily raise costs, disproportionately affecting low-income families. Furthermore, renewable energy solutions, like solar panels and heat pumps, require significant upfront investments that many disadvantaged households cannot afford (p. 3).

In The Irish NAP-DG, those in the private rental sector were identified as structurally vulnerable, and particularly those on lower income. It was noted that:

[T]his vulnerability also often intersects with other inequalities, for example age (young adults, students) as well as migrants who are renting from private landlords. These groups often rely on the private sector for accommodation, with high rents, particularly in urban settings. It has been noted that there is little incentive for private landlords to upgrade the energy rating of the buildings/apartments they own, thus leading to high energy bills for tenants (p. 5).

The Irish NAP-DG also mentions that “mechanisms in place (e.g. a grant system for retrofitting)” do not address the needs of low-income groups, including low-income people who are also single parents, parents of young children, have lower levels of education or have a migration background, also “in relation to their reliance on fossil fuels, and the energy inefficiency of their housing” (p. 5).

Older adults

Age is identified cross-nationally as a relevant ground of inequality in relation to the energy sector. In Ireland’s energy sector, older cohorts (60+) are affected, “particularly those living in houses which have not been retrofitted or upgraded to a higher energy efficiency rating” (IE NAP-DG, p. 5). In Spain, the elderly are also mentioned due to “systemic barriers in access to resources, employment, and technology” (ES NAP-DG, p. 4). In Italy, “older adults may have difficulty adapting to digital tools and technologies that are becoming more prevalent in the energy sector, such as smart meters or energy management systems” (IT NAP-DG, p. 4). This is because the user interfaces of these technologies are

not always designed with accessibility in mind. In Portugal and Hungary, elderly people on fixed pensions are particularly vulnerable to rising energy costs (PT NAP-DG, p. 4; HU NAP-DG, p. 4).

Women

Women are identified as a structurally disadvantaged group in Spain (ES NAP-DG, p. 4) and Italy (IT NAP-DG, p. 5). Other NAP-DGs mentioned gender in relation to its intersections with other grounds of inequality. See *Intersections of gender, age, income groups, care responsibilities and disabilities* below (p. 33).

Rural households

In Spain, it was noted that rural households “are more vulnerable to EP and to the digital gap, [which affect] their access to modern energy solutions” (ES NAP-DG, p. 4).

Similarly, in Italy:

[R]ural areas, particularly in the South [due to regional disparities], have less advanced infrastructure, which can limit access to the benefits of the digital energy transition. These regions may face challenges in integrating new digital technologies like smart grids, energy storage systems, or renewable energy sources. The digital divide in rural areas can mean limited access to high-speed internet, making it harder to participate in digital energy programmes or access real-time energy management tools. Additionally, rural areas often have a more centralised energy distribution system and lack the local, decentralised energy production models enabled by digital technologies, such as community solar or microgrids. (IT NAP-DG, p. 4).

In Hungary:

[R]ural and remote areas often have less developed energy infrastructure and limited access to natural gas and district heating, relying more on inefficient and polluting fuels (e.g., firewood, lignite). Access to renewable energy investments (such as solar and wind projects) is uneven, with wealthier regions benefiting more. Digitalisation of energy services (smart meters, online billing) may exclude people in low-connectivity areas (HU NAP-DG, p. 4).

In Portugal, “the green energy transition risks exacerbating urban-rural energy disparities if targeted investments are not made in rural energy infrastructure” (PT NAP-DG, p. 4).

Immigrant and minority communities

Immigrant and minority communities were mentioned in Italy, as “immigrant populations may be more vulnerable to EP and may have limited access to the digital tools needed for energy management”, with barriers including “language, low levels of digital literacy, and a lack of knowledge about available energy-saving or renewable energy programmes” (IT NAP-DG, pp. 4-5).

Roma people

The Italian case study notes that “minorities, including Roma communities, may live in areas with inadequate infrastructure, further limiting their access to the benefits of the digital energy transition” (IT NAP-DG, p. 5). Roma communities were also mentioned as structurally disadvantaged in other national case studies. In Ireland, members of the Traveller community and Roma community were

identified, and especially “those living in halting sites (Travellers) or in deprived accommodation in urban settings such as the North Inner City Dublin” (IE NAP-DG, p. 5).

In Hungary:

[M]any Roma families live in EP, often in segregated housing areas with poor insulation, inadequate heating, and unreliable access to electricity. High unemployment rates make it harder for Roma households to afford cleaner, more efficient energy sources. Discriminatory barriers can limit access to energy subsidies and state support programmes (HU NAP-DG, p. 4).

Similarly, in Portugal “Roma communities face systemic barriers to housing and energy access” (PT NAP-DG, p. 4).

Intersections of gender, age, income groups, care responsibilities and disabilities

An intersectional perspective is key in understanding EP. In Germany, starting from an intersectional understanding of structural vulnerability, gender and age are influential factors for the energy sector (DE NAP-DG, p. 4). The German NAP-DG reports that:

[T]he gender pension gap, housing conditions (living in older, non-renovated buildings) and the fact that women in the majority of cases live longer than men, and therefore often alone at an older age, puts especially older women at risk of living in EP. If we further look at the axis of ethnicity or migration history, a lot of women with migration history are living in non-renovated buildings oftentimes with several household members, that not all are contributing to the household income (kids, older people) (p. 4).

In the Irish case study it was noted that age can intersect with income groups, with “those relying on a state pension, or those on reduced pension (particularly women who were out of work due to caring responsibilities)” being especially vulnerable (IE NAP-DG, p. 5). In addition, people with a disability who are not in employment, as well as their carers who rely on social welfare, may be particularly vulnerable due to requiring specialist equipment, leading to higher costs for energy bills (p. 5). People with disabilities were also identified as structurally vulnerable in Spain (ES NAP-DG, p. 4) and Italy (IT NAP-DG, p.4).

Supply side

Workers in the fossil fuel industry and workers with low digital skills

Regarding the energy sector as a labour sector (supply side), in Germany workers from former coal regions are mentioned, “since they are directly affected by a lack of social compensation for transition measures and in need to adapt to new (digitalised) work cultures/tasks” (DE NAP-DG, p. 4). In Ireland, structurally disadvantaged workers include those in the peat industry (IE NAP-DG, p. 5). The Midlands is described as “a region which will require targeted interventions, and the local community is also targeted through the Irish Just Transition Fund” (p. 5). In addition, fossil fuel workers are affected “in industries where fossil fuels are being phased out” (p. 5). Lastly, “those who lack digital skills or those who lack ‘green’ skills in the energy sector and are in a need for upskilling” (p. 5). People who lost their jobs in coal phasing-out industries were also mentioned in Spain (ES NAP-DG, p. 4). In Italy, “workers in the fossil fuel industry, including those in coal, natural gas, and oil, are facing potential job

displacement as the energy sector transitions towards more sustainable and digital energy solutions” (IT NAP-DG, p. 5). In Portugal, “workers in the fossil fuel industry are threatened by the phase-out of [...] coal and oil refineries, which jeopardises their job security” (PT NAP-DG, p. 4). As job losses are concentrated in specific regions, this is likely to contribute to localised economic decline (p. 4).

4.1.2 Transport

Summary of DGs identified

Potentially disadvantaged groups		ES	HU
Demand side	Households headed by single women	X	
	Large families	X	
	People with disabilities	X	
	Older adults	X	X
	Women	X	
	Rural and insular households	X	X
	Low-income people	X	X
	LGBT+ people	X	
	Racialised people	X	
	Roma people		X
Supply side	Workers in the fossil fuel industry	X	
	Workers in pollution-intensive sectors		X

Table 4. Summary of DGs identified in the NAP-DGs in the Transport sector.

Demand side

In Spain, several groups have been identified in reports and surveys as the most affected by transport poverty in the twin transition. The most frequently mentioned groups include: “low-income people, unemployed people, single-women households, large families, people with disabilities and the elderly, as these are the groups most likely to experience difficulties affording public transportation or cleaner vehicles (also Electric vehicles - EVs)” (ES NAP-DG, p. 4). In addition, “people with disabilities, elderly

people and LGBTIQ+ individuals experience difficulties with accessibility and safety in public transport” (p. 4).

The Hungarian NAP-DG reports that “in Hungary’s transport sector, several structurally disadvantaged groups face inequalities due to infrastructure disparities, economic constraints, and policy gaps” (p. 5). DGs identified include people in poverty, who are likely to be more affected by rising fuel prices, vehicle maintenance costs and public transport fares (p. 5). Additionally, “many low-income individuals rely on public transportation, which may not always be affordable, frequent, or effective in connecting rural areas. [...] EVs and alternative transport solutions (e.g., bike-sharing, car-sharing) remain financially out of reach. [...] People living in rural areas are also at a disadvantage due to poor public transport connectivity, car dependency and limited infrastructure for green transport” (pp. 5-6).

The elderly population, identified as structurally disadvantaged in both Spain (ES NAP-DG, p. 4) and Hungary (HU NAP-DG, p. 6), “may face physical barriers in accessing public transport due to inadequate infrastructure (e.g., lack of elevators, ramps)” (HU NAP-DG, p. 6). In addition, “many modern transport systems rely on digital ticketing and ride-hailing apps, which some elderly individuals struggle to use (p. 6). Roma communities, identified as structurally disadvantaged in Hungary, sometimes “live in isolated settlements with poor or no public transport links, making access to jobs, healthcare, and education difficult. [...] High poverty rates limit access to private vehicles and alternative mobility solutions, [and] Roma individuals often report discriminatory treatment in public transport and workplace barriers in transport-related jobs” (p. 6).

Supply side

On the supply side, a structurally disadvantaged group identified in Spain is that of people who lost their jobs in coal phasing-out industries (ES NAP-DG, p. 4). In Hungary, people employed in pollution-intensive sectors (e.g., car manufacturing), were identified as structurally disadvantaged due to job insecurity (HU NAP-DG, p. 7). This is because “the shift towards EVs and automation in manufacturing threatens traditional car industry jobs in Hungary’s automotive sector, which employs over 170,000 people” (HU NAP-DG, p. 7).

4.1.3 Building & Housing

Summary of DGs identified

Potentially disadvantaged groups		DE	PT
Demand side	Low-income people	X	X
	Rural households	X	
	Children	X	
	Older adults	X	X

	Young adults		X
	Pregnant people	X	
	People with disabilities and/or chronic illnesses	X	X
	People experiencing homelessness	X	X
	People working outdoors	X	
	People renting/living in social housing		X
	Women-headed households/households headed by single mothers		X
	People with migrant background	X	X
	LGBT+ people		X

Table 5. Summary of DGs identified in the NAP-DGs in the Building & Housing sector.

Demand side

In relation to the Building & Housing sector, the “structurally vulnerable population groups taken into consideration by policymakers in Germany include children, older people and pregnant women as well as people with chronic illnesses or physical disabilities” (DE NAP-DG, p. 4).

It was noted that, in Portugal:

[W]omen-headed households, particularly single mothers, are [a] group that is disproportionately affected by housing inequalities. These households face higher financial constraints and greater difficulties in securing energy-efficient housing. The gender pay gap and caregiving responsibilities limit women’s ability to invest in homeownership or energy-saving measures, which further exacerbates their financial vulnerability (PT NAP-DG, p. 4).

In Germany, “socioeconomic vulnerability factors affect inhabitants of structurally weak and rural regions, homeless people and people working outdoors” (DE NAP-DG, p. 4).

Using the lens of EP, in Portugal challenges related to the twin transition and the Building & Housing sector relate to affordability and climate resilience, with some groups disproportionately affected. For example:

[T]enants in energy-inefficient housing are especially vulnerable. Many rental properties, particularly in cities like Lisbon and Porto, are poorly insulated and expensive to heat. Additionally, the rising cost of rent, exacerbated by the housing crisis, makes it even more difficult for low-income families to find affordable, energy-efficient homes. Energy efficiency policies primarily target homeowners, leaving renters at a disadvantage (PT NAP-DG, p. 4).

People experiencing homelessness were identified as a structurally vulnerable group in both Germany (DE NAP-DG, p. 4) and Portugal (PT NAP-DG, p. 4). In Portugal, “the lack of affordable housing forces many individuals into homelessness or precarious living conditions, where access to [...] heating and cooling is extremely limited” (PT NAP-DG, p.4).

In Germany, research shows that people perceived as migrants face racism on the housing market, having less ability to find affordable housing (see Salikutluk & Podkowik, 2024 p. 11). Consequently, their opportunities to save money for sustainable housing, i.e. related to renewable energy supplies, is limited, which can further enhance vulnerabilities (see for example Domann, 2016). Especially refugees have nearly no decision-making power over their living conditions, especially not about housing and energy supply” (DE NAP-DG, p. 5).

In Portugal, young workers and students were identified, “particularly those living in temporary, high-rent accommodations in urban areas, as they often face the combined challenge of poor energy efficiency in their housing and limited financial resources” (PT NAP-DG, p. 5), which makes them vulnerable to rising energy costs.

4.1.4 Food & Agriculture

Summary of DGs identified

Potentially disadvantaged groups		IE	IT
Demand side	Low-income households	X	X
Supply side	Migrant workers	X	X
	Workers with low digital literacy	X	
	Households with poor Internet connection	X	
	Women	X	X
	Rural farmers		X
	Carers	X	
	Young farmers	X	
	Older farmers	X	X
	New entrants to the sector	X	
	Small farmers/family-run farms	X	X

	Workers engaged in informal labour	X	
	Low-income farmers		X
	Farmers engaging in traditional practices/organic farmers		X

Table 6. Summary of DGs identified in the NAP-DGs in the Food & Agriculture sector.

Demand side

Considering consumers, in Ireland it was noted that socio-economically disadvantaged people may be at a structural disadvantage in the twin transition from a food poverty point of view. This is because “food poverty is an outcome of other factors such as unemployment or under-employment, housing inadequacy, homelessness, migrant status and fuel costs” (IE NAP-DG, p. 6). In Italy, structurally disadvantaged groups on the consumer side included consumers in low-income urban areas:

[W]hile the digital transition in agriculture might improve food quality and sustainability in the long run, low-income urban consumers may face higher prices, exacerbating food insecurity and access issues (IT NAP-DG, p. 7).

Supply side

Considering workers, in Ireland the groups identified include those living in areas with poor internet connections, women farmers, young farmers, new entrants to the sector, small farmers, and older farmers who lack digital skills (IE NAP-DG, p. 5). Similarly, structurally disadvantaged groups on the supply side in Italy were identified as small-scale farmers and family farms, farm workers especially including migrant workers, women in agriculture, rural communities and those living and working in remote areas, older farmers, farmers using traditional practices and those running organic farms (IT NAP-DG, pp. 6-7).

In Ireland:

[T]he agricultural and land-use system is made up of thousands of predominantly small family enterprises based in every part of the country. [...] Transition will principally rely upon thousands of individual farmers innovating and changing practices or land use, across diverse soils, land types and production systems. The active participation and engagement of farmers is therefore of fundamental importance to effective transition. [...] The challenges of enabling generational renewal, advancing gender equality, providing a living wage, and enabling the health and wellbeing of farm families all require a much better understanding of the farm family-farm relationship by policy makers. They entail dialogue and social consensus building, the development of a collective understanding of the challenges facing the sector, and the co-development of equitable solutions that enable the health and wellbeing of primary producers. The prioritisation of using only economic incentives to address social challenges has had limited success and impact. (IE NAP-DG, pp. 5-6).

In Italy, it was noted that inequalities “often stem from socio-economic, regional, and technological disparities that affect access to and participation in digital agriculture innovations” (IT NAP-DG, p. 6). Small-scale farmers and family-run farms, which are prevalent in Italy:

[O]ften lack the financial resources and technical expertise to invest in or implement digital tools such as precision farming technologies, automated irrigation systems, or data analytics platforms. These farmers may struggle with high upfront costs and may not see immediate returns, which can discourage them from adopting new technologies, thereby exacerbating inequality between large, industrialised farms and smaller, more traditional operations. Family farms may also be slower to adopt digital innovations due to generational gaps in digital literacy and resistance to change, which further widens the gap with more technologically advanced operations (IT NAP-DG, p. 6).

In addition:

Italy's rural areas, especially in the South, often face infrastructural challenges, including limited access to reliable high-speed internet and modern telecommunications, which are essential for adopting digital technologies in agriculture. Remote regions may also lack sufficient support services, such as training and technical assistance for digital tools, further exacerbating the gap between urban and rural areas in terms of digital agriculture adoption (IT NAP-DG, pp. 6-7).

Traditional farming practices may pose farmers at a disadvantage, "particularly in southern Italy and rural areas", as they "may not be well-suited for or easily integrated with modern digital technologies, especially if those technologies do not consider specific local knowledge or agricultural methods" (IT NAP-DG, p. 7). In addition, "farmers who focus on organic, sustainable, or agroecological practices may not always benefit from the digital transition, particularly if digital tools and innovations are designed with conventional agricultural practices in mind. The integration of new technologies into organic farming practices may be complicated or incompatible with sustainability goals, leading to the exclusion of this group from the digital benefits available to more conventional farm models" (IT NAP-DG, p. 8).

Migrant workers make up "a significant portion of Italy's agricultural labor force", and they are "often excluded from digital transformation efforts in the sector" (IT NAP-DG, p. 6). Women in the Italian agricultural sector may also be limited by structural disadvantages in benefitting from digital agriculture advancements (IT NAP-DG, p. 6).

One group that emerged (e.g., in Ireland) was that of people engaged in informal labour, "as they lack employment protection and are often on low incomes, while not being able to access social welfare support" (IE NAP-DG, p. 6). It was noted that:

[T]his category also frequently intersects with other inequality grounds, most likely migration status, in particular those who are undocumented and do not have access to regular work. Precarious workers are also potentially vulnerable due to the inconsistencies in incomes and reduced ability to access mortgages, thus relying on private rental accommodation. In addition, those working as unpaid carers looking after family members are also potentially vulnerable, especially if their access to social welfare is limited. Some studies also point out that those working as self-employed are at risk of EP, which is most likely to be linked with irregular or unpredictable incomes and limited access to welfare payments. (IE NAP-DG p. 6).

4.1.5 Cross-sectoral analysis

Cross-sectoral overview

Potentially disadvantaged groups		Energy	Transport	Building & Housing	Food & Agriculture
Demand side	Low-income households	X	X	X	X
	Older adults	X	X	X	
	Young people	X		X	
	Children			X	
	Women	X	X		
	Pregnant people			X	
	Rural households	X	X	X	
	People with migrant background/Racialised people	X	X	X	
	Roma people	X	X		
	People with disabilities/chronic illnesses	X	X	X	
	Carers	X			
	People not in employment	X			
	People renting/living in social housing	X		X	
	People with low level of education	X			
	Single parents	X			
	People living in older buildings	X			
	Households headed by single women		X	X	

Potentially disadvantaged groups		Energy	Transport	Building & Housing	Food & Agriculture
	Large families		X		
	LGBT+ people		X	X	
	People experiencing homelessness			X	
	People working outdoors			X	
Supply side	Workers in the fossil fuel industry	X	X		
	Workers in the peat industry	X			
	Workers with low digital literacy	X			X
	Inhabitants of former fossil fuel regions	X			
	Workers in pollution-intensive sectors		X		
	Migrant workers				X
	Households with poor Internet connection				X
	Women				X
	Rural farmers				X
	Carers				X
	Young farmers				X
	Older farmers				X
	New entrants to the Agriculture sector				X
	Small farmers/family-run farms				X
	Workers engaged in informal labour				X

Potentially disadvantaged groups		Energy	Transport	Building & Housing	Food & Agriculture
	Low-income farmers				X
	Farmers engaging in traditional practices/organic farmers				X

Table 7. Cross-sectoral overview of DGs identified in the NAP-DGs in the Transport sector.

Across all four sectors of the FITTER analysis, there is a broad distinction in structurally disadvantaged groups between those on the demand/consumers side and those on the supply/producers side of sectoral activity. Low income has been identified as a vulnerability consistently across sectors and case study countries.

On the demand side, variables intersecting with having a low-income are relevant across all sectors. The German case study noted that, across sectors:

[A]ge is an important factor for vulnerability due to a low pension in Germany as well as age-related physical conditions. This intersects with the category of gender: the gender pay gap and gender pension gap result in women having a lower socio-economic status and, as a consequence, less financial back-up. It also translates into decision-making power, e.g. regarding housing updates or energy supply, e.g. when it comes to property, deciding and financing for renewable energy technology or supply or for renovation measures. Furthermore, the housing and energy crises (in cities) affect mainly tenants in comparison to owners. In deregulated and profit-oriented housing markets, tenants are exposed to rent increases and the decisions of landlords or real estate agencies regarding the energy supply: hence they might be exposed to high gas and oil prices; or even if renewable energy measures such as heating pumps are installed, often tenants bear the costs of the modernisation levy (see e.g. Großmann 2019). Finally, physical conditions such as illnesses and disabilities and economic possibilities also overlap (DE NAP-DG, pp. 4-5).

There was an overlap between groups most frequently identified as potentially disadvantaged across multiple sectors in Ireland, Spain and Italy, which include low/income households, women (particularly those in low/income and/or single-parent households), people with disabilities, households in rural areas and low-income elderly people with low digital skills. Additionally, in Spain unemployed people were mentioned, and in Hungary the potential disadvantages faced by Roma communities was underlined.

In Spain, individuals (particularly women) engaged in informal labour and young people were mentioned, including those working in unpaid caregiving roles, particularly caring for older adults or people with disabilities (ES NAP-DG, p. 5).

In Ireland, migrants were mentioned because “with regard to racist incidents in the workplace ‘Ireland ranked as one of the worst Europe-wide with a rate of 33%, compared to an average of 24%’ (TheJournal, 2023). Many migrants are concentrated in low-skilled and low-paid jobs due to racial discrimination, language barriers, or lack of access to higher education. They also heavily rely on

privately rented accommodation and, in some cases, have limited access to social housing” (IE NAP-DG, p. 7). Migrants were also identified in Spain as a DG.

In Germany, “both the energy and housing sector reflect structural and complex intersectional inequalities related to chronic illness, disability and discrimination of LGBTQIA+ communities (see Habersbrunner et al. 2024; Großmann 2017)” (DE NAP-DG, p. 5). In Ireland, LGBTQIA+ communities face discrimination in accessing rental accommodation and accessing jobs, and can face discrimination or harassment on public transport (IE NAP-DG, pp. 6-7). In Spain, LGTBIQA+ individuals face systemic challenges due to “transphobic discourse and restrictive policies” (ES NAP-DG, p. 5), while in Portugal LGBTQ+ communities “may encounter additional barriers in accessing affordable, energy-efficient housing due to discrimination” (PT NAP-DG, p. 5). In Hungary, individuals whose interests are in conflict with the government’s position face many vulnerabilities, de to the erosion of the rule of law (HU NAP-DG, p. 7).

Disadvantage on the supply side was noted in all sectors with the exception of Building & Housing. In the Energy, Transport and Food & Agriculture sectors, groups of workers who may be disadvantaged by the twin transition include those in fossil fuel industries, car manufacturing (specifically internal combustion engine vehicle production), and small-scale farming. Across the case study countries, it was noted that disadvantage in these labour sectors is characterised by a strong regional dimension, with some regions more strongly affected than others. For example, in Germany workers from former coal regions are mentioned (DE NAP-DG, p. 4), while the Irish Midlands is identified as a region which will require targeted interventions (IE NAP-DG, p. 5). There is also a strong urban/rural divide: for example, in the Food & Agriculture sector, farmers living in areas with poor internet connection, older infrastructure or using traditional farming methods are more at risk of adverse effects from twin transition policy. The Food & Agriculture sector was the one where considerations on disadvantage on the supply side were strongest. In addition to small-scale and rural farmers, vulnerable groups identified in this sector include women farmers, young farmers, new entrants to the sector, older farmers, as well as migrant and seasonal workers.

Intersectional analysis

In many cases, the DGs identified in the NAP-DGs were specific, in that they noted disadvantage at the intersection of multiple grounds or domains of inequality. This table reports the specific groups identified listed according to the main grounds of inequality they are connected to. This table highlights the importance of using an intersectional approach.

	IE	DE	ES	IT	HU	PT
Gender	Agriculture and energy: Women on low income, single mothers Agriculture: Women farmers (less likely to inherit land and when they do inherit land, they tend to own smaller farms on poorer land)	Energy: Women with migration history living in non-renovated buildings with several household members not contributing to the household income (kids, older people)	Energy: Women not in employment and with low digital literacy			Housing: Single mothers, low-income women
Age	Energy: Older people living in houses which have not been retrofitted, relying on a state pension or reduced pension (e.g., women who were out of work)	Energy: Older people living in older, non-renovated buildings, especially women	Energy: Older people in rural areas	Energy: Older adults with low digital literacy	Energy: Older adults living in older, poorly insulated homes, and/or with low digital literacy Transport: Older adults with low digital literacy	Energy: Elderly individuals who live in older, poorly insulated homes Energy and housing: Young workers and students, particularly those living in temporary, high-rent

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
	Agriculture: Older farmers who lack digital skills Agriculture: Younger farmers (generational renewal is very slow - passing the farm from older to younger generation - often due to inadequate pension provision for older farmers - means younger farmers may be doing a lot of unrecognised work)					accommodations in urban areas
Ability	Energy: People with a disability who are not in employment					
Socioeconomic status	Energy: Socio-economically disadvantaged migrants; low-income people with low level of education	Energy and Building: Low income and disabilities/chronic illnesses	Energy and transport: Individuals (particularly women) engaged in informal labor and young people, including those working in unpaid caregiving roles	Agriculture: Low-income consumers in urban areas Energy and Agriculture: Low income persons,		Energy: Workers in affected sectors with low digital literacy

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
	Energy/agriculture: low-income people with low digital skills Energy: those in private rental sector on lower income, especially young adults/students and migrants Energy: parents of young children on low income Energy: single-parent families on low income Energy: Carers who have no other income and/or who require specialist equipment that raises energy bills			especially women, elderly people and migrants, and people leaving in rural communities Energy: workers in affected sectors with low digital literacy Agriculture: small, low-income farmers, women farmers, older farmers		

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
Race/Ethnicity	Agriculture: Migrant workers in affected industries Agriculture: Migrants workers in informal labour, with caring responsibilities and renting in the private market Energy: Roma people living in deprived accommodation in urban settings	Energy and Building: People with low economic possibilities facing racism in the housing market		Energy and Agriculture: Immigrant people with low digital literacy	Energy: Roma people in segregated housing areas with poor insulation, inadequate heating, and unreliable access to electricity; Roma people not in employment	Energy: Roma people living in temporary or informal housing, or in situations of economic instability
Sexual orientation		Energy and building and housing: LGBTQ+ people with chronic illness and/or disability				Energy and Housing: LGBTQ+ people with low income

Table 8. Overview of intersectional DGs identified in the NAP-DGs.

4.2 Intermediary organisations identified

4.2.1 Energy

In Ireland, organisations mapped relate to several of the groups identified as at-risk of EP in the context of the twin transition. Considering low-income groups these include Anti-Poverty Network Ireland, Co-operative Housing Ireland, Community Work Ireland, DePaul, Financial Justice Ireland, FLAC Free Legal Advice Centre, Focus Ireland, Peter McVerry Trust, Threshold Preventing Homelessness, An Cosan, Brighter Community Energy CLG. Organisations working with the elderly include University of the Third Age, Friends of the Elderly Ireland, Sage Advocacy and Age & Opportunity. Gender inequalities and injustices are the focus of the National Women's Council, Feminist Communities for Climate Justice Project, Rape Crisis Network Ireland and Women's Aid.

The Irish case study also identified additional research groups, NGOs, governmental organisations and agencies relevant to inequalities in the Energy sector. These include "the Economic and Social Research Institute (ESRI) who has been tasked by the government to conduct the analysis of EP in Ireland for the Energy Action Plan as well as several researchers based in UCD (UCD Energy Institute and UCD Geary Institute for Public Policy)" (IE NAP-DG, p. 12). NGOs include "Friends of the Earth Ireland, St. Vincents the Paul organisation, Threshold (for tenants affected by EP) and Social Justice Ireland" (p. 12), while governmental organisations include "National Economic and Social Council (NESC) with specific programme on energy transition, as well as the Irish Human Rights and Equality Commissioner (IHREC)" (p. 12).

The German case study has identified organisations working specifically on EP, such as Caritas, Verbraucherzentrale, and in particular on energy debt counselling (Verbraucherzentralen ausgewählter Länder, e.g. Berlin). Inhabitants of structurally weak and rural regions are addressed primarily by federal programmes (e.g., Deutscher LandFrauen Verband e.V.). Organisations who work with people experiencing homelessness include BAG Wohnungslosenhilfe e.V. and Wohnungslosenstiftung. ProAsyl e.V. and Bellevue di Monaco represent people seeking refuge, while organisations working with people with a migrant background include DaMigra e.V., Morgen e.V., Amadeu-Antonio-Stiftung and Netzwerk Rassismus- und Diskriminierungsfreies Bayern e.V.. Tenant associations include Deutscher Mieterbund e.V., Mieterverein München e.V., Mieterhilfe e.V.,_Mieter helfen Mietern e.V.,_and Verbraucherzentrale München. Youth organisations include Katholische Landjugend Bewegung and Arbeitsgemeinschaft der evangelischen Jugend, while relevant trade unions include ver.di and DGB München. There are also Religious institutions working on climate aspects (e.g. NourEnergy e.V., Ökumenisches Netzwerk für Klimagerechtigkeit, Jewish Youth Climate Movement) and public institutions working with people, who need support in socio-economic aspects (e.g. Jobcenter).

The Spanish case study has identified organisations that represent the interests of or conduct work relevant to low-income people (Madrid Foodbank - FESBAL, Formació i Treball, Fundación Atenea, Incubadora con Valores), the elderly (Amigos de los Mayores, Fundación Atenea), women (SINGA), rural households (Ecologistas en Acción), immigrant and minority communities (SINGA, Spanish Commission of Refugee Aid - CEAR), people with disabilities (Ilunión), and potentially disadvantaged workers (Workers' Commission/Comisiones Obreras - CCOO, Federación de Industria, Construcción y Agricultura - UGT FICA).

In the Italian case study, workers are represented by CGIL, CISL, UIL and ACLI. Women's rights are represented by Stati Generali delle Donne, while low-income communities and social inequalities are addressed by Forum Disuguaglianze e Diversità (also represents the interests of immigrant and minority groups). Rural and environmental issues are considered by Alleanza Italiana per lo Sviluppo Sostenibile, Legambiente and WWF Italia.

In the Hungarian case study, organisations focused on low-income groups include A Város Mindenkié (focus on homelessness), ADOM, Egységes Diákfront (focus on low-income students), Nem tehetsz róla, tehetsz ellene, aHang, Civil Kollégium Alapítvány, Alternatíva Alapítvány, Utcáról Lakásba Egyesület (focus on homelessness), Igazgyöngy Alapítvány, Tanítanék Mozgalom (focus on low-income students), Nyomtass Teis. Organisations focused on women include Magyar Európai Nők, Civil Bázis and Civilizáció Koalíció. UNICEF Hungary focuses on child welfare. Organisations that support elderly people include Nyugdíjasok Országos Képviselőlete - NYOK and Nyugdíjas Szervezetek Egyeztető Tanácsa - NYUSZET. Organisations who work with citizens living in rural communities include Települési Önkormányzatok Országos Szövetsége - TÖOSZ, Községek és Kistérségek Országos Önkormányzati Szövetsége - KÖSZ, Magyar Önkormányzatok Szövetsége - MÖSZ, Kisvárosi Önkormányzatok Országos Érdekszövetsége - KÖÖÉSZ, Magyar Gazdakörök és Gazdaszövetkezetek Szövetsége - MAGOSZ.

The Portuguese NAP-DG reports that "in Portugal's energy sector, several structurally disadvantaged groups face significant challenges that require focused attention and advocacy. Low-income households are particularly affected by EP, exacerbated by high energy costs and inefficient housing. Organisations like CASC - Centro de Apoio Social de Cascais and DECO - Associação de Defesa dos Direitos dos Consumidores work to represent and support these communities. The elderly population, many of whom live on fixed pensions and in poorly insulated homes, also struggle with high energy costs, making them another vulnerable group. Key stakeholders for this group include CNIS - Confederação Nacional das Associações de Reformados, Pensionistas e Idosos and Associação de Aposentados de Portugal. Rural and remote communities, facing energy infrastructure deficits, are also disadvantaged, with organisations such as FENASE - Federação Nacional das Associações de Desenvolvimento Local and AMRT Algarve - Associação de Municípios da Região de Turismo do Algarve working to address these issues. Migrant and Roma communities often live in substandard housing and encounter barriers to accessing energy, with organisations like ACM - Agência para a Imigração advocating for their rights. Workers in fossil fuel industries, who are impacted by job losses due to the transition from fossil fuels to renewable energy, are represented by trade unions like FIEQUIMETAL - Federação Intersindical das Indústrias Metalúrgicas, Químicas, Elétricas, Farmacêutica, Celulose, Papel, Gráfica, Imprensa, Energia e Minas and CGTP - Confederação Geral dos Trabalhadores Portugueses, which focus on protecting their interests during this transition" (p. 6).

Additional relevant organisations include Academia and research institutions, such as Instituto Superior Técnico - IST and Universidade do Porto, and employers and industry representatives in the renewable energy sector (APREN - Associação Portuguesa de Energias Renováveis) (PT NAP-DG, p. 7).

4.2.2 Transport

With reference to the groups identified in the previous section, some organisations have been identified in the Spanish and Hungarian case studies who represent the interest of those affected by the twin transition in the Transport sector.

The Spanish NAP-DG maps trade unions who may represent the interests of workers affected by the twin transition in the transport sector (e.g., Comisiones Obreras - CCOO, The Federation of Industry, Construction and Agriculture/Federación de Industria, Construcción y Agricultura - UGT FICA). In addition, consumer organisations have been mapped (Organisation of Consumers/Organización de Consumidores - OCU, Federación de Consumidores y Usuarios - CECU).

Considering organisations that interact with low-income cohorts, these include the Madrid Foodbank - FESBAL and Fundación Atenea, who carries out interventions with vulnerable groups at all stages (prevention, assistance, harm reduction, integration, etc.) but also in social research (own research, analysis and evaluation of public policies, formulation of social projects). Amigos de los Mayores fights loneliness in elderly cohorts, while several organisations work with migrants and refugees (e.g., SINGA - association that generates links and inclusive ecosystems between migrants, refugees and local people through entrepreneurship, creativity and design; Spanish Commission of Refugee Aid - CEAR). Ilunió advances social justice in the workplace for people with disabilities. More stakeholder organisations could be sought who support women, Roma communities, and queer communities.

In the Hungarian case study, organisations focused on low-income groups include A Város Mindenkié (focus on homelessness), ADOM, Egységes Diákfront (focus on low-income students), Nem tehetsz róla, tehetsz ellene, aHang, Civil Kollégium Alapítvány, Alternatíva Alapítvány, Utcáról Lakásba Egyesület (focus on homelessness), Igazgyöngy Alapítvány, Tanítanék Mozgalom (focus on low-income students), Nyomtass Te is! (focused on free publishing). Organisations focused on women include Magyar Európai Nők, Civil Bázis and Civilizáció Koalíció. UNICEF Hungary focuses on child welfare. Organisations that support elderly people include Nyugdíjasok Országos Képviselője - NYOK and Nyugdíjas Szervezetek Egyeztető Tanácsa - NYUSZET. Organisations who work with citizens living in rural communities include Települési Önkormányzatok Országos Szövetsége - TÖOSZ, Községek és Kistérségek Országos Önkormányzati Szövetsége - KÖSZ, Magyar Önkormányzatok Szövetsége - MÖSZ, Kisvárosi Önkormányzatok Országos Érdekszövetsége - KÖÖÉS, Magyar Gazdakörök és Gazdaszövetkezetek Szövetsége - MAGOSZ. Further stakeholder mapping could focus on organisations that represent people with disabilities, Roma communities, workers affected by the twin transition and queer communities.

4.2.3 Building & Housing

In the Portuguese Building & Housing sector, organisations identified in the NAP-DG include AIL - Associação Inquilinos Lisbonenses and Confederação Nacional de Habitação, vital stakeholders for advocating for tenant rights and improved housing conditions. The homeless population faces a lack of access to affordable and energy-efficient housing, with organisations like Associação Crescer and Santa Casa da Misericórdia playing crucial roles in providing support and solutions. Women-headed households are particularly vulnerable, often facing financial constraints and EP. Key stakeholders working on behalf of these households include CIG - Comissão para a Igualdade de Género and AML -

Associação de Mulheres de Lisboa, which focus on ensuring that gender-sensitive policies address their specific needs in the housing sector.

Some of the organisations listed in the German NAP-DG represent the interests of groups likely to be negatively affected by the twin transition in the Building & Housing sector. These include organisations who support socioeconomically vulnerable people such as tenants, migrants, women and youth. Organisations who work with people experiencing homelessness include BAG Wohnungslosenhilfe e.V. and Wohnungslosenstiftung. ProAsyl e.V. and Bellevue di Monaco represent people seeking refuge, while organisations working with people with a migrant background include DaMigra e.V., Morgen e.V. and Netzwerk Rassismus- und Diskriminierungsfreies Bayern e.V.. Tenant associations include Deutscher Mieterbund e.V., Mieterverein München e.V., Mieterhilfe e.V.,_Mieter helfen Mietern e.V.,and Verbraucherzentrale München. Organisations supporting queer communities access housing include Housing First Queer and QueerHome. Energy debt counselling organisations are relevant, as are youth organisations (Katholische Landjugend Bewegung, Arbeitsgemeinschaft der evangelischen Jugend).

4.2.4 Food & Agriculture

With reference to the groups identified in the previous section, some organisations have been identified in the Irish and Italian case studies who represent the interest of those affected by the twin transition in the Food & Agriculture sector.

In Ireland, several organisations were identified who represent the interests of migrants, potentially knowledgeable about challenges faced by migrant workers in the Food & Agriculture sector. These include the Irish Network Against Racism, the Irish Refugee Council, Doras, the Immigrant Council of Ireland, Movement for Asylum Seekers Ireland, Nasc - Migrant and Refugee Rights, the Irish Immigrant Council and Migrant Rights Centre Ireland. Considering small and/or organic farmers, the Irish organisation Talamh Beo focuses on land use tenure and agrarian change. People living in rural Ireland are represented by organisations such as Irish Rural Link, Irish Countrywomen's Association (also includes a gender dimension), Women in Agriculture Stakeholders Group, and Macra na Feirme, a community of 17–40-year-olds from 170 clubs in rural Ireland.

When considering additional contacts or organisations who may be relevant in the Food & Agriculture sector, the Irish case study listed trade unions such as SIPTU Services Industrial Professional Technical Union – Relevant Divisions; and Fórsa – Relevant Divisions. Academics researching inequalities in agriculture were also identified.

In Italy, organisations mapped with sectoral relevance are currently more high-level and potentially less in touch with members of potentially disadvantaged groups. These include environmental NGOs such as WWF Italia, Alleanza Italiana per lo Sviluppo Sostenibile, and Legambiente. More efforts are needed to identify organisations representing the interests of migrant workers, small farmers, women farmers, younger and older farmers, those working in isolated or rural areas, using traditional practices or running organic farms.

4.2.5 Overview

The table below presents an overview of the intermediary organisations identified by national case studies in the NAP-DGs (also listed sectorally in the previous sections), organised by DG and country.

	IE	DE	ES	IT	HU	PT
Low-income households	• Anti-Poverty Network Ireland • Co-operative Housing Ireland • Community Work Ireland • DePaul • Financial Justice Ireland • FLAC Free Legal Advice Centre • Focus Ireland • Peter McVerry Trust • Threshold Preventing Homelessness • An Cosan • Brighter Community Energy CLG	• Energy debt counseling (Verbraucherzentralen ausgewählter Länder, e.g. Berlin) • Caritas • Verbraucherzentrale • JobCenter	• Madrid Foodbank - FESBAL • Formació i Treball • Fundación Atenea • Incubadora con Valores	• Forum Disuguaglianze e Diversità	• ADOM • Egységes Diákfront (focus on low-income students) • Nem tehetsz róla, tehetsz ellene • aHang • Civil Kollégium Alapítvány • Alternativa Alapítvány • Igazgyöngy Alapítvány • Tanítanék Mozgalom (focus on low-income students) • Nyomtass Teis	• CASC - Centro de Apoio Social de Cascais • DECO - Associação de Defesa dos Direitos dos Consumidores
Older adults	• University of the Third Age • Friends of the Elderly Ireland		• Amigos de los Mayores • Fundación Atenea		• Nyugdíjasok Országos Képviselőte - NYOK	• CNIS - Confederação Nacional das

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
	- Sage Advocacy - Age & Opportunity - National Federation of Pensioners Association				Nyugdíjas Szervezetek Egyeztető Tanácsa - NYUSZET	Associações de Reformados Pensionistas e Idosos Associação de Aposentados de Portugal
Young people	- The National Youth Council of Ireland - Union of Students Ireland	- Katholische Landjugend Bewegung - Arbeitsgemeinschaft der evangelischen Jugend - Jewish Youth Climate Movement				
Children	Leadership for Inclusion in the Early Years -LINC				UNICEF Hungary	
Women	- National Women's Council - Feminist Communities for Climate Justice Project - Rape Crisis Network Ireland		SINGA	Stati Generali delle Donne	- Magyar Európai Nők - Civil Bázis - Civilizáció Koalíció	- Comissão para a Igualdade de Género - AML - Associação de Mulheres de Lisboa

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
	Women's Aid					
Rural households	Irish Rural Link	Deutscher LandFrauen Verband e.V.	Ecologistas en Acción	- Alleanza Italiana per lo Sviluppo Sostenibile - Legambiente - WWF Italia	- Települési Önkormányzatok Országos Szövetsége - TÖOSZ - Községek és Kistépülések Országos Önkormányzati Szövetsége - KÖSZ - Magyar Önkormányzatok Szövetsége -MÖSZ - Kisvárosi Önkormányzatok Országos Érdekszövetsége - KÖÖÉS - Magyar Gazdakörök és Gazdaszövetkezetek Szövetsége - MAGOSZ	- FENASE - Federação Nacional das Associações de Desenvolvimento Local - AMRT Algarve - Associação de Municípios da Região de Turismo do Algarve
People with migrant background	- Irish Network Against Racism - Irish Refugee Council	- ProAsyl e.V. - PRO ASYL – Der Einzelfall zählt	- SINGA - Spanish Commission of Refugee Aid - CEAR	Forum Disuguaglianze e Diversità		

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
/Racialised people	• Doras • Immigrant Council of Ireland • Movement for Asylum Seekers Ireland • Nasc – Migrant and Refugee Rights • Irish Immigrant Council • Migrants Rights Centre Ireland • Brazil for All • Association of Ukrainians in Ireland • Syrian Association of Ireland • New Communities Partnership	• Bellevue di Monaco • DaMigra e.V • Morgen e.V., • Netzwerk Rassismus und Diskriminierungsfreies Bayern e.V. - MORGEN München • Netzwerk Rassismus- und Diskriminierungsfreies Bayern e.V.,				
Roma people	• Pavee Point-Traveller and Roma Centre • Irish Traveller Movement					

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
People with disabilities/chronic illnesses	• AsIAm - Ireland's Autism Charity • Aware • Disability Federation Ireland • Enable Ireland • Inclusion Ireland • Irish Deaf Society • Irish Wheelchair Association • Mental Health Ireland • National Council for the Blind • Neurodiversity Ireland • Social Farming Ireland		• Ilunión			• Associação Portuguesa de Deficientes
People renting/living in social housing	• Treshold • Community Action Tenants Union	• Deutscher Mieterbund e.V. • Mieterhilfe e.V. • Mieter helfen Mietern e.V. • Verbraucherzentrale München				• Associação Inquilinos Lisbonenses) • Confederação Nacional de Habitação

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
LGBT+ people	• Belong To LGBT+ • Dublin Lesbian Line • LGBT Ireland • TENI Transgender Equality Network Ireland	• Queermany • Housing first Queer • QueerHome				
People experiencing homelessness	• Dublin Simons Community • Crosscare	• BAG Wohnungslosenhilfe e.V.: • Wohnungslosenstiftung • Housing first Queer • QueerHome			• A Város Mindenkié • Utcáról Lakásba Egyesület (focus on homelessness)	• Associação Crescer • Santa Casa da Misericórdia
Workers in affected sectors/industries	• SIPTU Services Industrial Professional Technical Union – Relevant Divisions; Fórsa – Relevant Divisions; Energy Services Union of Ireland - ESU • Unemployment Alliance Ireland	• Trade Unions: ver.di • DGB München	• Workers' Commission/Comisiones Obreras - CCOO • Federación de Industria, Construcción y Agricultura - UGT FICA	• CGIL • CISL • UIL • ACLI		• FIEQUIMETAL - Federação Intersindical das Indústrias Metalúrgicas, Químicas, Eléctricas, Farmacêutica, Celulose, Papel, Gráfica, Imprensa, Energia e Minas • CGTP - Confederação

D2.6 Methodological Guidelines for the Identification and Involvement of DGs and Action Plans
FITTER-EU

	IE	DE	ES	IT	HU	PT
						Geral dos Trabalhadores Portugueses
Women farmers	- Irish Countrywomen's Association - Women in Agriculture Stakeholders Group					
Rural farmers	Irish Farmer's Association (IFA)					
Young farmers	Macra na Feirme					
Small farmers/family-run farms	- Talamh Beo - Land Use, Tenure and Agrarian Change - ICMSA					

Table 9. Overview of intermediary organisations identified in the NAP-DGs by national case study and DG.

4.3 Summary of national contexts for Stakeholder Engagement: Comparative analysis of challenges and incentives

4.3.1 Germany

In pp. 8-9, the German NAP-DG reported potential challenges for the stakeholder engagement of DGs. The NAP-DG mentions that there may be differences in challenges related to different structurally disadvantaged groups, which should be considered in terms of “access to devices, different burdens and barriers specific social groups face, their economic backgrounds as well as time capacities” (DE NAP-DG, p. 8). In addition, the importance of avoiding box-ticking approaches, tokenism and victimisation is underlined, as it is key to include stakeholders in a meaningful way.

Challenges listed include differences in digital literacy (primarily disadvantaging older people, but also people with disabilities, and intersections with gender and migration background), which could be mitigated by “[offering] material as printed versions, [organising] small trainings on digital literacy in advance or [ensuring] support by assigning partners among intergenerational participants” (DE NAP-DG, p. 8). In addition, challenges may arise due language and/or literacy in relation to people with refugee or migration background whose first language is not German, which could be mitigated by translating materials into different languages. Other tools and media may be combined to engage illiterate people or people with disabilities (e.g., audio, large print, sign language or braille) (DE NAP-DG, p. 8).

The German NAP-DG also notes that “Physical lack of accessibility of in-person meetings could impact negatively on people with disabilities and chronic illnesses” (p. 8). The Plan also mentions offering and combining different modalities for participation (e.g. online – in person, synchronous – asynchronous) to meet the needs of those with (unpaid) care work responsibilities, often women, or engaged in precarious wage labour (p. 9). This could include, for instance, the organisation of online vs in-person meetings, as well as synchronous – asynchronous, including the recording of videos which might be consulted later in case a participant is not able to attend to a specific meeting.

Challenges mentioned also include the backlash against discussions of topics related to gender and denial of climate change, especially when engaging with people of different political orientations or “disconnected from current political debates due to their marginalisation” (p. 8). Providing information and structuring engagement in an inclusive way could support political awareness-building to gain trust and interest.

Incentives listed by WECF (p. 9) include offering gender trainings in exchange for participation, providing visibility and networking possibilities (e.g., on social media or via internal networking tools), and including participant perspectives in WECF’s political work. WECF considers remuneration for participants in the form of vouchers (e.g., for children’s products, groceries or public transport), or by partnering with CSOs who could offer something (e.g., “free language courses for non-German speakers or free training programmes to reduce economic and bureaucratic burdens”, p. 9). WECF will offer food and beverages in in-person meetings and will try to consider specific participants’ needs (e.g., in terms of childcare). Lastly, “WECF can offer to support with understanding energy bills, e.g. by supporting with getting heating costs back and give lessons on energy use and possible ways of energy saving” (p. 9).

4.3.2 Ireland

The Irish NAP-DG (p. 14) structures challenges and incentives for the stakeholder engagement of DGs in five broad categories: issues related to time, financial issues, trust and interest in research activities, language/literacy barriers, and accessibility issues.

Issues related to time are likely to affect participation in workshops for “those from lower socio-economic backgrounds as they may have multiple jobs or working shifts and irregular hours” and “parents of young children and those with other caring responsibilities” (p. 14). In addition, considering the Agriculture sector, “additional work (e.g. beside farm duties and often during 9-5 frame) should be accounted for” (p. 14). Possible mitigations include flexibility in scheduling (considering evenings and weekends to include various groups) and organising workshops closer to stakeholders, e.g. outside of Dublin for farmers.

Financial issues emerge as TU Dublin cannot offer financial incentives, especially for the participation of lower-income groups. However, refreshments/lunch may be budgeted for in-person activities, and reimbursing travel expenses could be considered. Otherwise, online participation should be offered.

In terms of trust and interest in participating, “trust and interest should be considered particularly in relation to migrant groups who may not be willing to participate due to their legal status and the fear of being exposed. Furthermore, research ‘fatigue’ can also be a barrier in relation to groups who are subject of various studies (e.g. Traveller community) as this is relatively small cohort and members may not be willing to participate as continuous research has not significantly improved their overall situation to date. Using civil society organisations as gatekeepers and recruiters may also pose challenges as the organisations may not be willing to provide details for potential participants due to privacy concerns.” (p. 14)

Mitigations include informing potential participants about the importance of their contribution and ensuring that they can express their views. In addition, “increased visibility for those who would like their issues to be highlighted is also a potential incentive to participate” (p. 15). Intermediary organisations could be asked to advertise the project and project participatory activities rather than identify potential participants. Additionally, “members of the research team should attend events organised by CSOs for networking and recruitment” (p. 15).

Language and literacy barriers relate in particular to groups with a migration background, but may also affect other groups due to the complexity of the topics to be discussed and a perception of certain topics as “too technical” (p. 15). As a result, “clear and simple language must be used in all communication” (p. 15) so that a high level of fluency in English will not be required. Other mitigations mentioned include peer translation/interpreting, or using innovative participatory methods such as using drawings.

Accessibility issues relate both to physical accessibility (of buildings, public transport, etc.) and digital accessibility (whether potential participants have the Internet connection, devices and skills required to join the activity).

4.3.3 Spain

The Spanish NAP-DG lists the following challenges for the engagement of structurally disadvantaged groups: accessibility and inclusion (physical, digital and linguistic), time and availability (due to “irregular work schedules, caregiving responsibilities, or mobility constraints”, p. 9), trust and willingness to participate, and complexity of topics. To address these, it will be key to ensure clear communication and ethical safeguards, as well as “ensuring that discussions and materials are accessible, avoiding excessive jargon or academic language, and providing clear explanations” (p. 9). While direct financial incentives cannot be offered, the following strategies and incentives are suggested (pp. 9-10): providing visibility by promoting the organisations who participate and disseminating their work, encouraging student involvement, highlighting the advantages of engagement, and fostering a safe space through compliance with ethical guidelines (including privacy, accessibility and flexible scheduling).

4.3.4 Italy

The Italian NAP-DG describes challenges related to a possible limited interest in participating in the project's activities, mainly in organisations which are usually involved and consulted in public policy participatory processes –usually related to the Cohesion Policy. This risk is minimised because “participation of Italian civil society is considerably limited in policy decision-making processes, particularly those related to the NRRP, which was characterised by a top-down decision-making structure, focused on ministerial bodies and regional and local bodies solely dedicated to implementation. In NRRP policy processes, CSOs, particularly those focused on structurally vulnerable groups, have been limited to specific contributions such as parliamentary hearings.

Furthermore, CSOs in Italy are experiencing a hostile environment that limits their field of action due to political actors who have discredited them and excluded them from decision-making spheres. This is exacerbated for CSOs working on immigration issues (Chiodi et al., 2021). Similarly, CSOs working on gender equality, sexual and reproductive rights, and the rights of LGBTIQ+ people are also facing such restrictions and hostility through exclusion” (p. 13).

Other challenges for the engagement with DGs in the Italian concerns the involvement of intermediary organisations accurately reflecting the needs of the structurally vulnerable groups they are representing. In particular, engagement should avoid replicating the disparities between the North and the South of the country. Other identified challenges are digital access and language barriers in the case of migrant population –one of the most vulnerable groups in the country–.

Among the identified incentives that SV and FGB could provide to promote stakeholder engagement, there is offering visibility and facilitating “between involved organisations and other stakeholders, such as policymakers—particularly at regional and local levels. FGB has extensive contacts with policymakers, trade unions, and researchers across all levels of government. SV has strong experience and well-established networks with stakeholders working on gender issues, innovation, and inclusion” (p.14). SV also identified the possibility of offering capacity-building online and in-presence sessions, as well as dedicated training sessions for participatory organisations. Finally, participants special or accessibility needs will be collected and some services to assure inclusiveness will be provided, such

as translation or childcare if the financial resources allow it. In addition, travel costs and catering expenses will be covered by the partners organising in-presence events.

4.3.5 Hungary

The Hungarian NAP-DG (p. 10) describes challenges and incentives for the engagement of structurally disadvantaged groups by emphasising that the process is “critically hindered by the lack of direct financial incentives, which significantly impacts participation” (p. 10). This is because reimbursement would be a “key cornerstone of the citizen’s assembly methodology” (p. 10), especially for the purpose of involving people in poverty, those living in remote areas and Roma communities, who face “real financial and logistical barriers” (p. 10). Digital exclusion is also a challenge for the involvement of low-income and elderly populations.

SFC could offer transportation support using company vehicles or partnering with local transport services. Food and refreshments can also be offered.

4.3.6 Portugal

The Portuguese NAP-DG (pp. 8-9) identifies the following challenges for the engagement of structurally disadvantaged groups: economic barriers, digital literacy and digital access, language and cultural barriers, trust, health and mobility issues, and institutional barriers.

The following incentives are listed to support participation (pp. 9-10):

- Reimbursement of transportation costs;
- Childcare support;
- Training and educational opportunities;
- Gift cards or vouchers;
- Access to energy efficiency resources;
- Food and refreshments;
- Peer support or mentorship.

4.3.7 Common challenges

	IE	DE	ES	IT	HU	PT
Digital literacy and access	X	X		X	X	X
Language and literacy barriers	X	X	X	X		X
Physical accessibility and mobility issues	X	X	X			X
Time and availability	X	X	X	X		X
Financial barriers	X			X	X	X
Trust and willingness to participate	X		X	X		X
Complexity of topics	X	X	X			X

Table 10. Challenges identified in the NAP-DGs by National Case Study.

4.3.8 Common incentives

	IE	DE	ES	IT	HU	PT
Financial support	X	X		X	X	X
Visibility and networking opportunities	X	X	X	X		X
Childcare support		X		X		X
Educational opportunities		X		X		X
Food/refreshments	X	X		X	X	X
Accessibility and inclusivity	X	X	X	X	X	X

Table 11. Incentives identified in the NAP-DGs by National Case Study.

5 Guidelines for the engagement of structurally vulnerable groups in the FITTER-EU project

Literature focused on the engagement of structurally vulnerable groups identifies multiple challenges when implementing participatory activities and initiatives with people affected by conditions of disadvantage. It also addresses incentives to ensure their meaningful participation, avoiding the reproduction of existing inequalities and reflecting their real needs. Section 3.1.2 categorises these challenges as follows: (i) challenges related to the limitations in resources that participants face in ensuring their participation, which depend on funding, time, and personnel costs; (ii) challenges in adequately identifying and recruiting participants (initiation challenges); and (iii) difficulties in ensuring actual collaboration, particularly in establishing ongoing cooperation that truly reflects participants' needs and does not perpetuate conditions of vulnerability (Amann & Sleight, 2021) (Mulvale et al., 2019a).

During the implementation of the FITTER-EU project, resource-related issues have emerged repeatedly, particularly the lack of a dedicated budget to support the various activities outlined in its participatory approach. In particular, time constraints, limited availability, and financial barriers have been reported as challenges in five and four of the NAPs-DGs, respectively. Ireland, Italy, and Portugal reported both types of barriers, while Germany and Spain noted only time-related constraints, and Hungary only financial ones.

Regarding recruitment and collaboration with DGs, partners in the NAPs pointed to several risk factors: limited digital literacy and access, language and literacy barriers, physical accessibility and mobility issues, low levels of trust or willingness to engage, and the complexity of the topics involved. Notably, partners from Portugal and Ireland mentioned all these challenges as relevant in their national contexts.

Further obstacles described in the literature include low self-confidence or perceived lack of expertise among participants, fear of stigma, and conflict avoidance, all of which can inhibit engagement. Additionally, power imbalances—whether between researchers, facilitators, and organisers, or among participants themselves due to social or economic differences—may further obstruct inclusive participation. A more detailed discussion of these issues is available in Section 3.1.2.

The active participation of structurally vulnerable groups in the FITTER-EU methodology begins in WP3 with a series of semi-structured interviews with individuals belonging to DGs foreseen in T3.3. Following, in WP4, tasks T4.1 “Scenario-based identification and analysis of inequality risks” and T4.2 “Co-creating intersectional mitigation plans” will also engage with structurally vulnerable groups. According to the GA, T4.1 involves deploying a questionnaire to assess the probability and severity of potential hazards, while T4.2 focuses on establishing national labs to jointly design mitigation strategies addressing the negative effects of the twin transition.

Although the effective roll-out of WP4 is still under discussion and some methodological aspects remain unresolved, this chapter offers practical guidelines to support the engagement of DGs within the FITTER-EU project.

5.1 FITTER-EU strategy for the engagement of structurally vulnerable groups: the role of intermediary organisations

Most of the challenges previously outlined in the literature and by FITTER-EU partners relate to the engagement of individual persons in participatory activities. To partially address these, the involvement of DGs in the FITTER-EU project is proposed to take place through intermediary organisations that represent them, rather than directly with individuals negatively affected by the twin transition. It is anticipated that working with such organisations may help overcome several of the barriers to participation encountered by vulnerable groups. In particular, intermediary organisations are likely to have greater resources in terms of time and personnel, thereby potentially reducing participation constraints.

Moreover, collaborating with structured organisations could enhance the co-creation of representative mitigation measures, as these entities are better placed to articulate the needs and experiences of the broader group they represent—beyond the perspectives of individual participants. In addition, engagement with intermediary organisations may help mitigate power imbalances within participatory settings, as these organisations often have experience interacting with diverse stakeholders and taking part in political or policy-related processes.

Engaging intermediary organisations appears particularly valuable for supporting the participatory activities under WP4. Specifically, Task 4.2, which involves the implementation of co-creation labs, will engage 15–20 participants. As the input from this relatively small group will be central to the development of proposed mitigation measures in the FITTER project, ensuring a wide variety of diverse perspectives inside each DG targeted by the project is essential. To this end, gathering the needs and proposals of a broader population—beyond the experiences of individual participants—is crucial. To enhance diversity within the labs, participation of DGs in T4.2 is proposed exclusively through members of organisations representing structurally vulnerable groups—the prioritised DGs by the project. Beyond the aforementioned benefits, it is worth considering that involving individuals not affiliated with organisations may require special consideration regarding data management.

T3.3 consists in carrying out interviews with individuals belonging to DGs, while T4.1 involves conducting a survey with individuals to assess the probability and severity of negative impacts. Unlike the FITTER labs, these two tasks necessitate the direct involvement of individual participants. However, even in these cases, intermediary organisations—if engaged and interested in the FITTER-EU project—could serve as valuable entry points to reach a wider group of stakeholders, thereby increasing the likelihood of achieving a representative sample for the survey in T4.1 and a more diverse set of answers in T3.3.

Engagement of intermediary organisations in participatory approaches with DGs is not without precedents: in health (co-creation) interventions, proxies are used when participation is deemed to be too challenging for individuals with vulnerabilities (Islind et al., 2023b). In another example, when looking at citizen engagement in sustainability transitions research, Huttunen et al. (2022) found that out of the 62 examined articles, in 10 it was reported that citizens were represented by NGOs instead of their direct participation. Even in the context of co-creation under the base-of-the-pyramid

approach, some found engaging with only NGOs sufficient to develop representative solutions (in the example, business models) that (see examples listed by Nahi, 2016).

Following, a series of benefits and drawbacks related to the engagement of intermediary organisations are detailed.

Benefits

- Vulnerable groups may be too fragile or unable to represent themselves and participation might be too challenging for them due to, for instance, illness or disability (Gesierich et al., 2024);
 - Simply switching to online settings is often inadequate to address this issue and (in certain cases it could in fact contribute to it, for example, in the case of elderly individuals with low digital skills) (Isind et al., 2023b).
- A significant risk when it comes to engaging with vulnerable groups is their overburdening and “over-using” (Kiernan & McMahon, 2024), and/or causing harm to them by involving them in the activities (OECD, 2024, p. 60) – relying on intermediaries might offer a workaround to that;
 - Often, vulnerable groups are “time poor” (OECD, 2024, p. 60), and the co-creation process can last for a long time; getting to know users and understanding their needs through “extensive interviews, co-design workshops and similar activities may not always be the most effective solution” (Isind et al., 2023b).
- Vulnerable groups’ participation is more difficult to secure and maintain (e.g., Hodson et al., 2023b; Mulvale et al., 2019b) as due to their vulnerable situations, they often have a more pronounced focus on day-to-day concerns and may not see the immediate value in participating (Mulvale et al., 2019b). Intermediaries may prove to be more consistent when it comes to participation;
- Vulnerable groups’ “genuine inclusion” is more difficult to reach and requires considerable effort and care, implying the need to develop specific facilitation competences;
- As intermediaries usually possess extensive knowledge and expertise on issues affecting the groups whose interests they aggregate and represent, their contribution might result in more meaningful results (at least when it comes to FITTER’s structure and objectives);
- If they have proper understanding of vulnerable groups’ experiences (Gesierich et al., 2024), they can be able to represent their needs (contact with gatekeepers, see Nahi, 2018);
- There are also potential benefits to the intermediaries, such as resulting in capacity-building benefits for the organisation and posing an opportunity for gaining valuable insights (Gesierich et al., 2024). There are also networking opportunities, as other potential incentives which has been mentioned by partners.

Drawbacks

- If DGs are only represented by their intermediaries, genuine inclusion/participation where participants are not only research subjects but also contributors to the design of a service/outcome (Hodson et al., 2023b) becomes impossible.
 - It is the default situation that DGs are overlooked because of the assumption of not having the capacity to participate fully in co-creation (Hodson et al., 2023b); yet again,

no “voice” would be given to those most severely impacted (Aldridge, 2015, p. 16), as somebody else would speak on their behalf.

- Even when intermediaries are knowledgeable, they often lack the lived experiences of vulnerable individuals. Hodson et al., p. (2023b, p. 271) argue that intermediaries “present limited, biased perspectives which cannot replace the voice of the vulnerable user”;
- While NGO representatives have extensive experience and knowledge, they may not realise that it’s relevant (or the extent of the relevance) (Nahi, 2018);
- Although engaging with intermediaries would probably be less resource-intensive, securing participation and issues of genuine inclusion are not exclusive to vulnerable groups:
 - For example, NGO-business relations may prove difficult to coordinate due to sectoral and cultural differences, different objectives and vocabularies (Nahi, 2018);
 - Power differentials would need to be paid maximum attention to in either case and facilitators should be trained equally well;
 - Barriers to participation, e.g., digital literacy issues may affect intermediaries as well;
- The difficulties of securing/maintaining participation of vulnerable groups have alternative solutions, e.g. over-recruiting (Mulvale et al., 2019b).
- Götzmann et al. (2020) emphasise (in a human rights-violation context) that sometimes it is difficult to identify legitimate representatives of rights-holders, as although an organisation might claim to represent a certain group, it may well be that they do not represent the opinions of the community members.

Most of the limitations associated with the engagement of intermediary organisations relate to the adequate representativeness of the vulnerable population they are intended to represent. In particular, intermediaries might not reflect the real diversity within the group, nor their needs, and may perpetuate inequalities by failing to allow vulnerable groups to “express their voice”.

As a consequence, it is fundamental to engage relevant and meaningful intermediary organisations that represent the interests and realities of their respective DGs. Efforts are therefore needed to identify and engage valid organisations in the WP4 participatory tasks of the FITTER project.

Following this, some questions are proposed which might help in their identification (adapted from Kiernan & McMahon, 2024, p. 15)

- Does the NGO (representative) have the participants’ best interest as a priority?
- Does the NGO (representative) have full understanding of the needs and vulnerabilities of the participant?
- Can such an alternative provide the same answers?

The following section provides an analysis of the organisations identified in the NAPs and some guidelines dedicated to enhancing the identification and mapping of vulnerable stakeholders, while the last part of the chapter provides some specific recommendations for the engagement of DGs in WP4 participatory tasks.

5.2 Recommendations to improve the mapping of intermediary organisations

A key step in the proposed strategy for involving DGs in project participatory activities is to ensure that the stakeholder organisations mapped in the Stakeholders Database accurately reflect the DGs identified in the project's theoretical deliverables to date, and in the NAP-DGs.

5.2.1 Gaps between theoretical deliverables and the groups mapped in NAP-DGs

Energy

The project's theoretical deliverables and initial findings note the importance of energy, digital and financial literacy in shaping people's ability to make cost-saving decisions in relation to their energy consumption (e.g., D2.1; T3.2). However, people with low levels of education, low digital literacy or low energy literacy were not explicitly mentioned in the majority of NAP-DGs. These aspects are partially linked to other variables, such as age and migrant background (e.g., due to proficiency in the local language), which may play a role in people's ability to access and make decisions based on information available. Further attention could be given to identifying groups with lower energy literacy and the organisations that represent them.

While several NAP-DGs noted the challenges faced by rural households with regard to EP, they did not focus on urban households, which the theoretical deliverables identify as being at higher risk of EP (D2.1).

Other groups which were identified as at-risk of EP in the project so far but were not explored to a great degree in the NAP-DGs include racialised people and ethnic minorities, socially isolated people and those living alone, large and multi-occupancy households, and people living in specific disadvantaged regions.

Box 2. DGs in Energy: Key action points for further mapping

- **ALL NATIONAL CASE STUDIES:** Map organisations that represent the interests of/work with DGs **in urban areas**, who the literature suggests are at greater risk of EP.
- **SPANISH AND ITALIAN CASE STUDIES:** Map organisations that represent the interests of/work with racialised people and ethnic minorities, socially isolated people and those living alone, large and multi-occupancy households, and people living in specific disadvantaged regions.
- **HUNGARIAN AND PORTUGUESE CASE STUDIES:** Map organisations that represent the interests of/work with racialised people and ethnic minorities and people living in specific disadvantaged regions.

Transport

Similarly to the Energy sector, in Transport early project findings highlight the importance of having the knowledge, understanding and skills to effectively engage with both public transport ticketing systems (e.g., having sufficient language and digital skills), as well as with possible grants and/or bureaucratic processes that may be in place to incentivise the purchase of electro vehicles. These

aspects were not extensively considered in the NAP-DGs, and relevant national case studies could benefit from deepening their consideration of these aspects (considered so far in relation to age).

The geographical/regional component is relevant in this sector as well, and further work could focus on identifying organisations that represent people from disadvantaged rural and suburban areas. Other groups identified in the theoretical deliverables and initial project findings not considered in the NAP-DGs include single-parent families, youth and full-time students and unemployed people.

If possible, it would be beneficial to reach specific subsets of workers who are likely to be experiencing Transport Poverty, such as those in the informal labour market and those working non-standard hours.

Considering the groups identified in the Spanish NAP-DG, in Spain more stakeholder organisations could be sought who support women, Roma communities, and queer communities. Similarly, in Hungary, further stakeholder mapping could focus on organisations that represent people with disabilities, Roma communities, workers affected by the twin transition and queer communities.

Box 3. DGs in Transport: Key action points for further mapping

- **SPANISH AND HUNGARIAN NATIONAL CASE STUDIES:** Map organisations that represent the interests of/work with disadvantaged rural and suburban areas, single-parent families, youth, workers who are likely to be experiencing Transport Poverty, such as those in the informal labour market and/or working non-standard hours, Roma communities, and queer communities.
- **SPANISH NATIONAL CASE STUDY:** Map organisations that represent the interests of/work with women.
- **HUNGARIAN NATIONAL CASE STUDY:** Map organisations that represent the interests of/work with people with disabilities and workers affected by the twin transition.

Building & Housing

Interview findings from T3.2 and ongoing case study work underlines the importance of bureaucratic understanding and literacy, as in this sector knowledge of grants and subsidies can be crucial (especially for homeowners). Digital skills also play an important, related role. While both NAP-DGs in this sector identify older adults as a DG, further work could consider other groups facing barriers in this regard, including racialised minorities and non-native speakers of the local language.

According to D3.1, “women who have experienced domestic violence are highly vulnerable to housing insecurity” (p. 71).

It would be beneficial to consider and map organisations who work in the interest of victims of gender-based violence.

Box 4. DGs in Building & Housing: Key action points for further mapping

- **GERMAN AND PORTUGUESE CASE STUDIES:** Map organisations that represent the interests of/work with victims of gender-based violence, who are vulnerable to housing insecurity.
- **PORTUGUESE CASE STUDY:** Map organisations that represent the interests of/work with racialised minorities and non-native speakers of the local language.

Food & Agriculture

There is considerable overlap between the DGs identified in the project's theoretical deliverables and early findings in the Food & Agriculture sector, and those identified in the relevant NAP-DGs, mainly covering the supply side. Further work could investigate more detailed breakdowns of consumers who may be at a disadvantage due to twin transition policy in Food & Agriculture, beyond low-income groups. For instance, specific household compositions may be at a disadvantage.

D2.2 identifies specific Food & Agriculture sub-sectors which may be at a disadvantage due to a change in dietary standards (e.g., meat production). The relevant national case studies could map the organisations (e.g., employers' organisations) who represent the interests of specific categories of producers viewed as vulnerable.

In the Italian case study, more efforts are needed to identify organisations representing the interests of migrant workers, small farmers, women farmers, younger and older farmers, those working in isolated or rural areas, using traditional practices or running organic farms.

Box 5. DGs in Food & Agriculture: Key action points for further mapping

- **ITALIAN AND IRISH CASE STUDIES:** Map organisations (e.g., employers' organisations) that represent the interests of specific categories of producers viewed as vulnerable
- **ITALIAN CASE STUDY:** Map organisations that represent the interests of/work with migrant workers, small farmers, women farmers, younger and older farmers, those working in isolated or rural areas, using traditional practices or running organic farms.

5.2.2 Gaps between Stakeholders Database and the groups mapped in NAPs

An analysis of the Stakeholders' Database (as of 03/04/2025) was conducted to assess the status of the project's mapping of organisations representing DGs, and whether it reflects the groups identified in the project so far.

In total, 319 organisations have been mapped which are reported to represent the interests of, or focus on, a DG. However, so far, the distribution of these organisations is not balanced across countries (Figure 1).

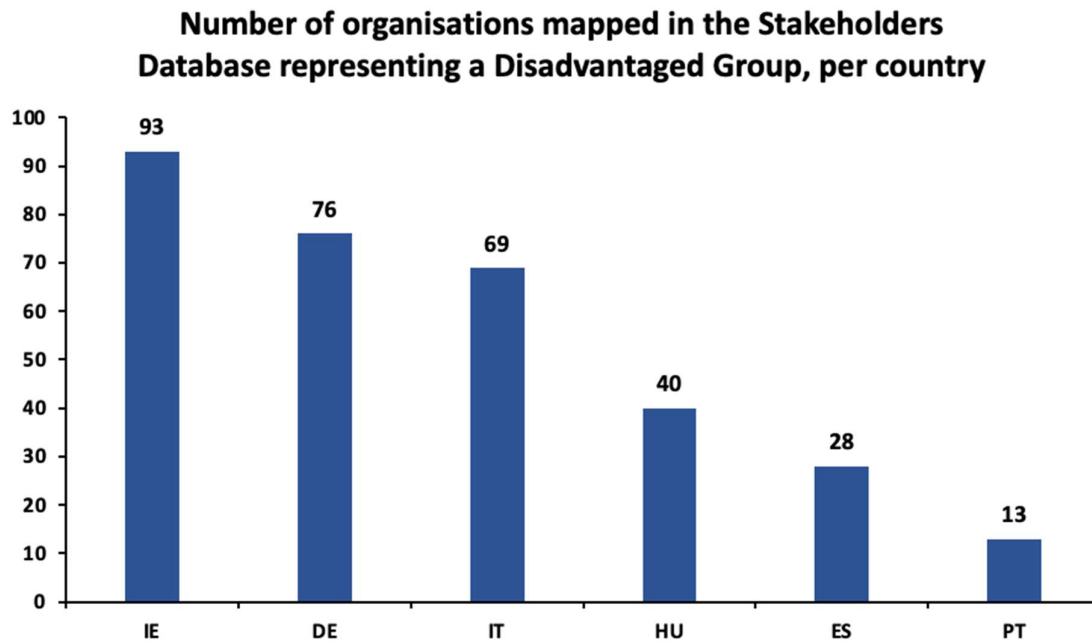


Figure 1. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per country

The organisations mapped which represent the interests of or focus on a DG are predominantly CSOs (182 organisations mapped), although public service/policymaking organisations (45), research/academic organisations (40) and trade unions (22) are also mapped. Unsurprisingly, considerably fewer business angels (3), employers' organisations (6) and private sector organisations (8) are reported to focus on DGs. This distribution varies considerably across countries, with Italy mapping the highest share of public service/policymaking organisations with a focus on DGs, and Hungary the highest share of trade unions (Table 12).

	Civil society organisation	Public service/policymaking	Research/academia	Trade union	Private sector	Employers' organisation	Business angel	Other
IE	67	11	14					1
DE	49	7	9	3	3			5
IT	19	21	12	5	3	3	3	2
HU	23	2	3	12				
ES	14	3	2	2	2	3		2
PT	10	1						2

Table 12. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per country and type of organisation

Considering sectoral relevance, there is a fairly equal distribution across the four sectors (Figure 2). It is worth noting that some of the organisations are focused on more than one sector; therefore, the total number of organisations included in the graph are higher than the total 319 organisations prioritising DGs.

Number of organisations mapped in the Stakeholders Database
representing a Disadvantaged Group, by sector*

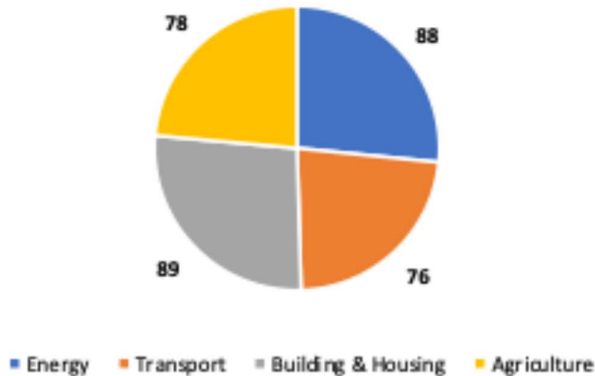


Figure 2. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per sector. *Some organisations are reported as relating to more than one sector.

Considering the groups represented, most of the organisations mapped are reported as representing the interests of or focusing on more than one DG (75). Beyond these, the vulnerabilities that are most represented in the database currently are poverty (50), gender (37) and disability/health impairments (27) (Figure 3).

Number of organisations mapped in the Stakeholders Database
representing a Disadvantaged Group, per group represented

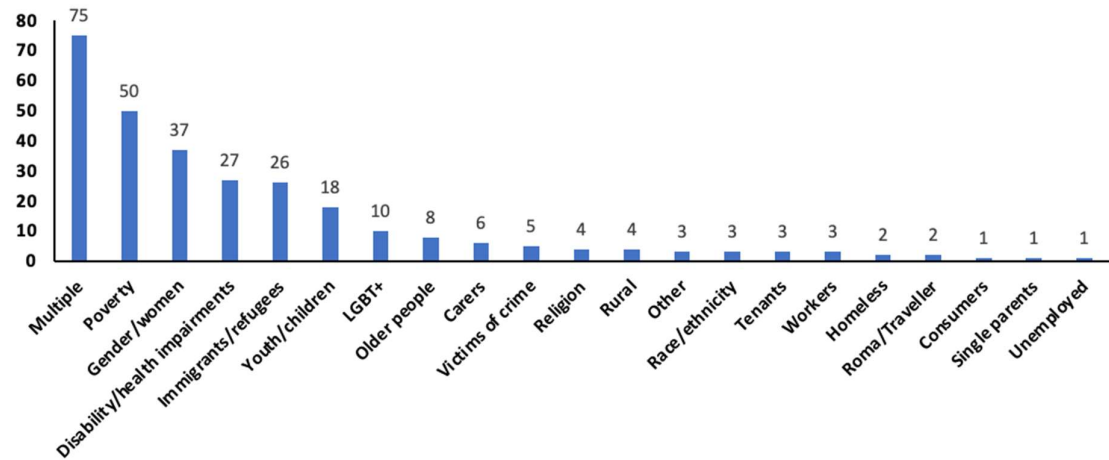


Figure 3. Number of organisations mapped in the Stakeholders Database reported to represent the interests of or focus on a DG, per DG

Relative to the DGs identified in the project’s theoretical deliverables, ongoing case study work, initial findings, and NAP-DGs, the range of organisations mapped in the Stakeholders Database largely focus on only a few of the groups identified. Further work could focus on mapping organisations that explicitly represent the interests of groups widely mentioned in the project’s research so far, but that

are not currently particularly represented in the Database, such as rural households, racialised people and ethnic minorities, or tenants.

Box 6. DGs identified in the Stakeholders database: Key action points for further mapping

- **SPANISH, ITALIAN, HUNGARIAN AND PORTUGUESE CASE STUDIES:** Map tenant associations, as people who do not own their home have been recurrently highlighted as potentially disadvantaged in several sectors; Map organisations that represent the interests of/work with rural communities.
- **GERMAN, SPANISH, ITALIAN, HUNGARIAN AND PORTUGUESE CASE STUDIES:** Map organisations that represent the interests of/work with older people.

5.3 Guidelines for WP4 participatory activities

WP4 participatory activities are concentrated in two of its four tasks: “T4.1: Scenario-based identification and analysis of inequality risks” and “T4.2: Co-creating intersectional mitigation plans”. The participatory character of both tasks differs significantly in terms of the type of stakeholders involved. The former will engage a broad number of individual participants to complete a survey aimed at calculating the probability of hazards and the severity of their negative impacts. On the other hand, the FITTER-EU labs will involve intermediary organisations representing DGs.

Following some general and specific recommendations per task are detailed.

General recommendations for engagement

- When starting the engagement: Tailor the message to the needs and relevance of the DGs, being respectful of their needs and clearly communicating expectations, as well as the potential benefits of participating in the project’s participatory activities.
- Consider the multiple existing challenges to engaging DGs and the high number of participants required for WP4 participatory activities; therefore, maximise efforts to map a broad number of relevant stakeholders and intermediary organisations. In the latter case, it is important to carefully evaluate whether they are valid organisations for the activities, i.e., whether they adequately represent the needs and opinions of the DGs they represent.
- “Start early”: It is recommended to establish initial contact with a broad number of intermediary organisations, as there is a high probability of refusals and potential dropouts throughout the process. Consider that over-recruiting might be needed with participants belonging to the most affected structurally vulnerable cohorts.
- Consider that some stakeholders belonging to structurally disadvantaged groups or intermediary organisations representing them might not be willing to participate in the FITTER-EU project initiatives due to their political affiliation which might differ from the FITTER-EU approach to fight against structural inequalities. This might be the case, for instance, of stakeholders affiliated with far-right organisations.
- It is advisable to offer the possibility of an initial informational meeting to address any inquiries or concerns, understand the DGs’ expectations and needs, and explain the process

of the participatory activity. This initial meeting could be held online or by phone if there are digital constraints.

- Even if the initial meeting does not take place, it is still recommended to collect the needs, expectations, and concerns of the DG's intermediary organisations before starting the participatory activities. This can be done with a questionnaire sent to expected participants.
- Based on this analysis, FITTER-EU partner organisations could assess how to support these needs as part of the available incentives for participation.
- Realistically consider the incentives—detailed in chapter 4.2—that partner organisations could offer to the DGs to promote participation, presenting them in a concrete and feasible manner after verifying with the organisation to avoid overpromising or creating false expectations.
- Possible economic incentives to encourage participation include vouchers, coupons, or shopping cards.
- Although the project does not have a specific budget for participatory activities, partners are strongly encouraged to evaluate their budget allocation to identify possible savings in travel and other direct costs that could be redirected to finance WP4 participatory activities. At the consortium level, an analysis is being carried out to identify possible savings which might allow the organisation of in-person FITTER-lab meetings, covering expenses such as venue rental, catering services, reimbursing participants' travel costs or the financing of other specialised services –such as braille or sign language interpretation– to guarantee accessibility with illiterate people or with special needs.

Engagement in T4.1

The FITTER-EU GA outlines the implementation of surveys per sector, aimed at collecting hazard probability rates and impact severity rates. Specific details regarding WP4 methodology are still under discussion at the project consortium level. However, there is agreement on the need to collect responses from individual participants. As part of the task, the sample size required to ensure 95% confidence and 80% power will be calculated, and the required number of participants will be equally distributed among the partners in the six project case studies.

Given the large number of stakeholders expected to be necessary for the sample size, it is recommended that partners engage with intermediary organisations representing DGs, which may assist in establishing contact with the targeted individuals.

When formulating the research questions, it is important to consider the information gap DGs may face with the research themes of the project. Therefore, to ensure a meaningful contribution from questionnaire respondents, the following considerations are recommended:

- Use plain language when formulating the questions, avoiding technical or specialist terms, and favouring accessible phrasing for non-specialist respondents.
- Contextualise the questions to the specific national and regional or local contexts of the respondents.
- Illustrate the questions with examples to which respondents can relate based on their personal experiences of being in a disadvantaged situation.

- Given the expected high number of respondents, it is likely that the questionnaire format will be online. A potential shortcoming of this approach is the exclusion of DGs with limited digital accessibility. If a targeted DG is affected by digital literacy or access issues, it may be necessary to implement part of the questionnaire in an in-person format using paper-based surveys.

Engagement in T4.2

Engagement in T4.2 is aimed at the development of the FITTER-EU labs, which, as previously mentioned, will target intermediary organisations. The labs have been conceptualised as a series of five meetings, as described in the GA:

1. Introductory in-presence workshop with DG representatives belonging to intermediary organisations to discuss the most relevant effects of the twin transitions and how the structurally disadvantaged groups they represent are affected.
2. One online working session between the project's partners and policy makers and academics part of the CCGS to identify a set of policy interventions (initial draft of mitigation measures)
3. In-presence workshop to present to the DGs representatives –intermediary organisations that participated to the first introductory workshop– the proposed mitigation measures to collect their opinions and feedback about potential solutions
4. Online session with policymakers and academics to refine and adjust the mitigation measures.
5. A final in-presence workshop with all the participants in the co-creation process to validate with the intermediary organisation representing DGs the new version of mitigation measures and conduct final adjustments of the proposal.

Given the existing budget constraints, a mixed online-in-person format is required. In-person meetings will be prioritised for engagement with DGs representatives –intermediary organisations– to ensure meaningful involvement and overcome challenges related to digital literacy. Thus, meetings 1 and 3 of the co-creation cycle will be held in person. This format may improve communication and foster trust within the group, as well as support the building of trust between the FITTER-EU researchers and the intermediary organisations representing DGs. It is worth considering that in some cases, members of intermediary organisations representing DGs might be part of the structurally cohort their organisation represent themselves. On the other hand, the second and fourth meetings dedicated to the proposal of mitigation measures and held with policy makers and academic experts will be organised online. The final event, involving intermediary organisations representing DGs, policy makers, and researchers, is also recommended to be in-person.

A stable group of participants throughout the process is preferred over involving different intermediary organisations, particularly in the sessions involving DGs, to ensure consistency in the proposal of mitigation measures. To minimise the risk of drop-out, it is important to consider the aforementioned recommendations. On the other hand, online sessions involving policy makers may involve different experts if they express time constraints regarding their participation. A key element to consider is that interaction with DGs requires specific attention to their needs; therefore, it is recommended that FITTER-EU partners involved in the labs have facilitation skills or receive training in this area.

Out of the six types of stakeholders targeted by the FITTER project, intermediary organisations representing vulnerable are most likely to be CSOs and trade unions². These stakeholders may be willing to participate in the project activities if their mission, vision, and activities align with counteracting the negative effects of the twin transitions. In addition, if they do not follow a specific thematic approach, their openness to participate may depend on how the project and co-creation activities are presented as potentially beneficial for their targeted group.

It is worth considering that many of these organisations adopt critical or anti-categorical approaches and often hold critical positions towards institutions and traditional politics. This could pose a challenge for meaningful engagement of DGs if the focus is on trade unions and larger, more institutionalised CSOs that traditionally participate in national policy processes. Therefore, it is essential to value the potential engagement of grassroots organisations or social movements that represent DGs not covered by institutionalised organisations. Engaging these stakeholders may require additional effort, but their involvement could enrich the development of mitigation measures and propose innovative solutions.

Power dynamics are another important factor to consider in the labs. Facilitators should ensure that all voices are heard during the session, moderating participants who tend to dominate the discussion and encouraging the participation of all involved. It may also be beneficial to establish collective ground rules at the start of the session and use facilitation techniques that promote direct interaction without hierarchies, such as alternating workgroups and plenary sessions, and encouraging feedback and constructive criticism from stakeholders. It is also important to value all types of contributions. Smart Venice will organise training sessions dedicated to partners who will be facilitating the FITTER labs.

Other recommendations include:

- Before the session, send participants a survey to collect information on their specific accessibility needs. Based on the responses, decisions may be made regarding the physical accessibility of venues, the provision of translators for language barriers, or the availability of childcare facilities.
- Similarly to the recommendations for the formulation of T4.1's survey, to ensure full understanding of the theoretical content, it is recommended to illustrate the content with relatable examples from the experiences of DGs. Plain language should be preferred over technical jargon.
- Referring to section 6.1.2 of D2.3 may be helpful in establishing engagement strategies based on the classification of stakeholders according to Mendelow's matrix. For instance, stakeholders with high influence and high interest in the project could be considered "key partners" and should have a central role in the labs. Stakeholders classified as "influencers," with low interest but high relevance to the project, are more susceptible to stakeholder

² The other types of stakeholders prioritised by the FITTER-EU project are: employers' organisations, public service/policymaking, private sector, research/academia, and business angels.

fatigue. Therefore, their involvement could be prioritised in only one of the sessions, such as the first lab, to collect their needs. If these stakeholders are unavailable for subsequent labs, they could provide feedback in an online format or through written exchanges in later stages of the process.

- Relevant stakeholders who have expressed explicit interest in contributing to the project through the bottom-up stakeholder engagement approach or in previous participatory activities should be contacted first for the development of WP4 activities.

A final consideration regarding the engagement of DGs is the inclusiveness checks proposed as part of the project's general engagement strategy, detailed in D2.5. Inclusiveness checks consist of two activities. First, SV will provide feedback in dedicated sessions analysing the stakeholders mapped and involved in the participatory activities. The first feedback session will be organised in May 2024, where specific recommendations will be provided to partners. Second, a self-evaluation questionnaire will be implemented after each participatory activity to encourage reflection by partners on the inclusivity of the sessions. After each lab session, the self-evaluation should be implemented, and necessary adjustments made if improvements are needed to enhance inclusivity.

The following box summarises the most relevant recommendations for the engagement of WP4 participatory tasks.

Box 7. Key recommendations for stakeholder engagement of WP4 participatory activities.

WP4 has two participatory tasks:

- T4.1 involving individual participants belonging to DGs through a survey to calculate the probability and severity rates of twin transition's negative impacts
- T4.2 involving intermediary organisations representing DGs, policy makers and academic experts in co-creation labs for the proposal of mitigation measures.

Recommendations for the engagement

- Tailor the message according to the expectations of the DGs, respecting their needs and collecting their expectations
- Start early: maximise your efforts for mapping intermediary organisations representing DGs, considering only "valid organisations" that represent the needs and opinions of the DGs. Over-recruiting might be needed
- Collect the needs and concerns of DGs with an initial informational meeting or with a questionnaire sent to participants before the start of the participatory activities.
- Prioritise plain language, contextualise the information and make relatable examples to the DGs' experiences.
- Partners are strongly encouraged to evaluate their budget allocation to identify possible savings in travel and other direct costs that could be redirected to finance WP4 participatory activities.
- When implementing T4.1 questionnaire consider the possibility to implement part of the questionnaire in an in-person format using paper-based surveys for targeted DGs with limited digital accessibility.
- For the FITTER labs in T4.2:
 - A stable group of participants throughout the sessions is preferred.
 - Sessions with intermediary organisations prioritising DGs will be held in-person.

- Facilitation should ensure inclusive, balanced participation. SV will organise training sessions (date to be defined).

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7 Annexes

7.1 Annex 1: National Action Plans for the Identification and Engagement of DGs

The template with the guidelines provided to partners for the NAPs-DGs can be found in the following [link](#).

They are included in the project's repository in the following path: "Documents/General/WP2 – Conceptual framework and methodology / T2.4 Engagement strategy for disadvantaged & vulnerable groups / National action plans for D2.6"

National action plans

- [Germany](#)
- [Hungary](#)
- [Ireland](#)
- [Italy](#)
- [Portugal](#)
- [Spain](#)

7.2 Annex 2: Preliminary identification of DGs

7.2.1 DGs identified in T2.1

Existing project deliverables (D2.1, D2.2, D3.1) have explored potential vulnerabilities uncovered and exacerbated by the twin transition.

D2.2 points out that there are gaps in knowledge on the societal impacts of the twin transition (Faggian et al., 2024), but key cultural norms, systems and structures that generate inequalities have been identified. These include relatively low wages in sectors such as healthcare, education, agriculture, and retail; gendered norms of care work (e.g. women do the majority of unpaid care work and are thus more likely to be in part-time and/or low pay roles); policy mechanisms which lead to the marginalisation or exclusion of key cohorts from educational courses and the labour market (e.g. people with disabilities, low skilled migrants, older people); inadequate housing supply for low income people; inadequate levels of rural connectivity; and inadequate social welfare supports. The resulting patterns of inequality can be identified in line with demographic categories such as gender, age, migration background, income levels, disability and geographical factors (e.g. regional differences, degree of urbanisation).

The digital divide is identified as a particularly strong indicator for vulnerability in the context of the digital transition, not least because of the interrelation of the digital and green transition. This is because digitalisation is widely perceived as a requirement for the green transition, for example in relation to sectoral measures such as installing photovoltaic panels or switching to an EV. The digital divide operates both in terms of access to the Internet, and in terms of inequalities in digital skills (D2.1). It affects a broad range of demographics, such as "older people, people with disabilities, low-income people, women, ethnic minorities, indigenous communities, displaced migrants and refugees"

(D2.1, pp. 35-36). The most documented structural disadvantages concern older people, whose difficulties in adapting to digitalisation have led to the term *grey digital divide* (D2.1, p. 36). In addition, Eurofound (2023) notes that inequalities in basic digital skills also exist between regions. Digital skill levels vary both between EU countries (with Southern and Eastern Europe at a disadvantage) and within them, along urban-rural divides (Feurich et al., 2024).

Strong regional differences exist in relation to structural vulnerability associated with the green transition. The ENTRANCES project designates several European regions as “Coal and Carbon Intensive Regions” (ENTRANCES, 2023, p. 5), often poorer regions in remote areas, which are likely to be greatly affected due to their reliance on a single industry, resulting in a cluster of challenges related to an aging population, limited job prospects, and difficulties in retaining young people (D2.1). An European Committee of the Regions Report points to regions in Czechia, Germany, Hungary and Poland which are already feeling the consequences of a loss of carbon-intensive jobs and a lack of green jobs to replace them (European Committee of the Regions, 2023).

As a result, several dimensions of inequality were identified in T2.1 deliverables as relevant in the context of the twin transition. The sub-chapter below presents sectoral vulnerabilities identified in the four sectors of analysis (Energy, Transport, Building & Housing and Food & Agriculture).

Sectoral vulnerabilities associated with the twin transition

Energy

In the energy sector, D2.1 reports structural inequalities (p. 44), which relate to broader barriers to women’s access to employment in the energy sector and gender stereotypes influencing horizontal segregation in secondary and tertiary education (D2.2, p. 15). Besides gender, different migration backgrounds and levels of education are not represented equally in this sector. This results in decisions within the sector being made by a largely homogenous group of people, which risks not reflecting the needs of all population groups. The green transition in the energy sector is a double/edged sword: while it will be essential in addressing and reducing EP, it may exacerbate temporary disadvantage for specific groups, especially considering that lower-income households usually spend a higher share of income on bills. These are: low-income households, people from ethnic minorities and indigenous people, immigrants and those with low proficiency in the local language, older people, disabled persons, women, young people and full-time students, single-parent families, socially isolated people and those living alone, as well as large households and multi occupancy households (D2.1, p. 45).

The intersections of these and other factors emphasise special inequality, for instance in terms of socioeconomic status, race and neighbourhood factors. An intersectional lens is key: single-parent households, which are more likely to anticipate difficulties with paying bills, are more often led by women, often with an ethnic minority background and low income (D2.1).

D2.2 underlines the importance of age and health in the context of EP. This is because illness and EP may go hand in hand in a vicious cycle, considering that illness can lead to poverty as more heating is required when staying indoors, but also that poverty can lead to illness. In addition, migration background is mentioned, as language barriers interfere with communicating with energy providers.

D2.2 points to intersectional inequalities linked with severe energy deprivation, encompassing low income, single parent households with a migrant background who need to spend many hours indoors.

Some national case studies have highlighted the role of geographical differences in determining EP and difficulties accessing renewable energy sources (for instance, in the Italian case, those living on islands, rural and suburban areas, or in the Southern regions) (D2.1).

Transport

Similarly to the Energy sector, employment in Transport and Mobility presents considerable imbalance in terms of gender and age, resulting in important inequalities in decision-making (D2.1).

Considering the users of transport services, availability and accessibility has important consequences for people with a disability. A useful concept is *time poverty*, as gendered travel patterns result in disadvantages for women, who often need to take multiple, short trips due to household and caring responsibilities (D2.2). There is a considerable overlap between the groups affected by EP and transport poverty, including people with low income, people with health conditions and disabilities, single-parent households, older people, women, and ethnic minorities; however, people in urban settings are more likely to experience EP, while people in rural and low-income suburban settings are more likely to experience transport poverty (D2.1). The link between the two phenomena is referred to as *double energy vulnerability* (D2.2).

The benefits of walking and cycling are not distributed evenly across the population (D2.1). In addition, men in urban areas with a high income, high levels of education and full-time employment, whose household already owns another car, are more likely to be owners of EVs (electric vehicles) than any other cohort (D2.2). As a result, penalising fossil fuel vehicles is likely to affect groups already at risk of transport poverty (D2.1), especially in rural areas and in suburbs with low-income households.

Specific professional profiles identified as vulnerable include car mechanics, logistics professionals and airport staff (D2.1).

Building & Housing

Employment in the Building & Housing sector is characterised by the underrepresentation of women and young people (D2.1).

DGs identified include tenants in private rental accommodation or social housing, which often overall with lower-income households (D2.1), due to the high likelihood of them being excluded from improvements to their houses' energy efficiency. There is also a gender impact, with women being more negatively affected. Renovation processes may negatively affect "historically marginalised groups, such as working class, low-income, racialised minorities, youth and older residents" (D2.1, p. 42). People with a disability may be further marginalised by smart city infrastructures.

Turning to those employed in the sector, there is a high rate of men and self-employed workers (often in precarious working conditions). In Italy, migrants and those working in the informal labour market are particularly vulnerable. Lastly, "low-skilled workers in construction could be disadvantaged by automation and technological progress" (D2.1, p. 44).

Food & Agriculture

The Food & Agriculture sector differs from the others, in that the affected population is primarily composed of workers in the sector, rather than end-users or consumers (D3.1). In particular, this applies to small and poorer farms, young farmers and women in agriculture (D2.2), due to the costs associated with change. The agricultural workforce belongs predominantly to older age brackets (D2.1). Regional divides play a key role (D2.2), while on the side of consumers, people with a lower socio-economic background are more likely to be negatively affected (D2.2).

7.2.2 DGs identified in T3.1

T3.1, and its findings as reported in D3.1, present the baseline: identified groups that are already disadvantaged in the context of each sector of analysis, and therefore their vulnerability in the context of twin transition policy should be researched, evaluated and understood. As a result, the focus is on existing vulnerabilities to EP, transport poverty, and disadvantage in relation to Building & Housing and Food & Agriculture, and whether these barriers can prevent these groups from fully participating in the shift to more sustainable energy, transport, housing or access to food, highlighting them as potential DGs in the twin transition context.

Energy

In D3.1, vulnerability in the Energy sector is researched in relation to EP.

The EP phenomenon is related to a series of demographic determinants, including age, gender, income, employment status, level of education and place of residence. These factors influence households' vulnerability to high energy costs and inadequate access to energy services. Furthermore, housing conditions and the energy efficiency levels of buildings are important indicators of a potential EP. The intersectionality of determinants that shape EP is particularly relevant as low-income households, the elderly, single parents, and rural populations are usually the most affected.

Income Levels

- Low-income households are the most affected by EP, as they allocate a higher percentage of their income to energy costs.
- Rising energy prices and economic instability (e.g., inflation, financial crises) further increase their vulnerability.
- The cost-of-living crisis in Europe, exacerbated by the COVID-19 pandemic and geopolitical conflicts (e.g., Russia-Ukraine war), has disproportionately impacted lower-income groups.

Employment Status and Job Security

- Unemployment or precarious employment increases the risk of EP, as these groups struggle with unstable income and high energy costs.
- Retired individuals on low pensions often face difficulty maintaining adequate heating or cooling in their homes.
- EP is often linked to seasonal work or low-wage jobs, particularly in sectors with high job insecurity. Housing Conditions and Energy Efficiency

- Poorly insulated homes or outdated heating systems increase energy consumption, leading to high energy bills.
- Social housing and rental properties are more likely to have low energy efficiency, as tenants often lack control over home improvements.
- In Eastern and Southern Europe, many older buildings lack proper insulation, contributing to higher rates of EP.

Education and Awareness

- Households with lower educational attainment often have limited access to energy-saving information and financial assistance programmes.
- Awareness of energy efficiency measures and government support programmes can help reduce EP, but information gaps persist in many regions.

Age

- Elderly individuals are particularly vulnerable due to fixed incomes, health issues, and reduced mobility, making it harder to adapt to energy shortages.
- Children in low-income households suffer from EP, impacting their education, health, and wellbeing.

Disability and health

- People with disabilities and people with chronic illnesses require stable indoor temperatures for their health (requiring higher energy consumption), making them more vulnerable to EP.

Gender

- Single-parent households, particularly those led by women, are at a higher risk due to lower income levels and caregiving responsibilities.
- Women, especially elderly women living alone, experience higher EP rates due to longer life expectancy and lower pensions.

Regional Disparities and the Urban-Rural Divide

- Rural areas often face higher EP rates due to lower incomes and fewer job opportunities, limited access to energy-efficient infrastructure and higher reliance on fossil fuels like coal or wood for heating.
- Urban areas also experience EP, particularly in low-income neighbourhoods with poorly maintained housing. (D3.1, pp. 30-31).

Transport

In D3.1, vulnerability in the Transport sector is researched in relation to Transport Poverty (TP).

Transport poverty is related to a series of demographic determinants, including age, gender, income, employment status, and ethnicity. These factors significantly influence transport mobility choices and can play a critical role in shaping transport accessibility and equity.

Age

- **Elderly People:** Older adults may face mobility issues due to physical disabilities, limited driving capabilities, lack of digital skills or inadequate public transport options tailored to their needs. Adoption of digital ticketing or smart mobility solutions may be challenging.
- **Youth and Students:** Young people, particularly those in rural areas, may struggle with limited or unaffordable transport options, restricting access to education and social participation.

Gender

- **Safety Concerns:** Women are more likely to experience safety-related issues while using public transport, affecting their willingness to travel at certain times or locations.
- **Trip Patterns:** Women often engage in more complex travel patterns, balancing employment, childcare, and household responsibilities, which may not be adequately supported by existing transport infrastructures.
- **Income Disparities:** Lower average earnings for women can make transport costs a higher burden.

Income

- **Affordability Barriers:** Low-income individuals often struggle with the high cost of transport, limiting their ability to access jobs, healthcare, and social services.
- **Geographic Disparities:** People in economically deprived areas may have fewer transport options, with inadequate investment in infrastructure and services.
- **Transport Choices:** Limited financial resources may force individuals to rely on inefficient or unsafe transport modes, increasing travel time and economic stress.

Employment Status

- **Job Accessibility:** Lack of affordable and reliable transport options can prevent unemployed individuals from seeking and maintaining employment.
- **Shift Work and Transport Availability:** Those working non-standard hours (e.g., night shifts) may face limited public transport availability, forcing dependence on costly or impractical alternatives.

Informal Employment: Workers in informal or unstable jobs may struggle with unpredictable transport costs and availability.

Ethnicity

- **Discrimination and Social Exclusion:** Ethnic minorities may face direct or indirect discrimination in transport services, reducing their access to equitable mobility solutions.
- **Language and Information Barriers:** Non-native speakers may have difficulty understanding transport information, schedules, and ticketing systems.
- **Residential Segregation:** Ethnic minority communities often reside in peripheral urban areas with poor public transport connectivity, exacerbating transport poverty. (D3.1, pp. 53-54).

Building & Housing

In Europe, housing vulnerabilities are closely linked to various demographic factors, such as age, gender, income, and other social determinants. These factors influence the risk of experiencing housing poverty, insecurity, and exclusion. Below are some key demographic vulnerabilities related to housing:

Age

- **Children and young people:** Families with children, particularly single-parent households, are at higher risk of living in poor-quality or overcrowded housing. Housing affordability issues affect young people as well, with many struggling to access housing due to low wages and high rental costs.
- **Elderly people:** Older adults, particularly those on fixed pensions, often face challenges in accessing affordable housing. Many are at risk of living in inadequate housing conditions, especially in cities where rents and housing prices have increased significantly. Elderly people may also face difficulties with accessibility (e.g., lack of elevators, poor maintenance of properties) and may experience social isolation because of housing instability.

Gender

- **Women:** Gender disparities in income and employment often lead to women being more vulnerable to housing poverty. Women are more likely to live in poverty, particularly single mothers or women over 65. Women are also more likely to face discrimination in the housing market, especially when dealing with issues related to child custody or domestic violence. Single mothers are in particular at risk of housing insecurity due to the challenges of balancing low wages with childcare responsibilities.
- **Gender-based violence victims:** Women who have experienced domestic violence are highly vulnerable to housing insecurity. They may need to leave abusive situations but struggle to find affordable housing due to financial constraints and a lack of specialised housing services for victims.

Income

- **Low-income households:** Low-income households are the most vulnerable to housing poverty. These families often spend a large proportion of their income on housing costs, leading to financial strain.
- **Youth and the unemployed:** Young people, especially those entering the workforce with low paying or insecure jobs, face challenges in securing affordable housing. Similarly, individuals experiencing long-term unemployment are at high risk of housing insecurity, particularly as housing costs outpace income growth. In the EU and the Euro area, the seasonally adjusted youth unemployment rate (under 25 years) has trended upward from November 2022 to November 2024. Over the past year, the EU's youth unemployment rate increased from 14.8 percent in November 2023 to 15.3 percent in November 2024.

Disability and Health Status

- **People with disabilities:** Disabled individuals often face additional barriers when it comes to accessing affordable and suitable housing. Inaccessible housing (e.g., lack of ramps, inadequate wheelchair access) and high costs make it particularly difficult for people with disabilities to secure adequate housing. Furthermore, disabled individuals often have higher living costs due to the need for specialised equipment or care.
- **Chronic health conditions:** Those with chronic health conditions, particularly when combined with low income or age, are more likely to live in substandard housing conditions that exacerbate their health problems. Poor-quality housing, such as inadequate heating or dampness, can have significant negative effects on individuals with respiratory issues or other chronic conditions.

Ethnic minorities and migrants

- **Immigrants and refugees:** Migrants, particularly refugees or undocumented individuals, face heightened vulnerability in the housing market. Language barriers, discrimination, and legal status can hinder their access to safe and affordable housing. Additionally, many migrants live in overcrowded conditions or are forced to live in informal housing arrangements due to limited access to the formal housing market.
- **Ethnic minorities:** People from ethnic minority backgrounds often face discrimination in the housing market, which can limit their access to quality housing. In some cases, these individuals may be subject to housing segregation or live in poorer neighbourhoods with higher levels of poverty and insecurity.

Geographic location

- **Urban vs. Rural:** People living in cities are often more vulnerable to housing poverty due to high rents and limited affordable housing. However, people living in rural areas may also face challenges, including a lack of affordable housing options and limited access to public services and employment opportunities. These factors create a unique set of vulnerabilities depending on the geographic location of individuals and families. (D3.1, pp. 70-73).

Food & Agriculture

The agri-food sector in Europe faces several demographic vulnerabilities related to age, gender, and income. A significant challenge is the aging workforce, with the average age of farmers increasing (across Europe, the number of farmers over 65 is notably high). This creates concerns about generational gaps in the sector and the need to attract younger individuals to farming. Indeed, only 11% of EU farm holdings are managed by individuals under 40 (Niranjan, 2024a, 2024b). Gender disparities are another issue, with women underrepresented in certain agricultural roles. For example, rural women in Europe are less likely to own a mobile phone compared to rural men, indicating significant technology access disparities. Moreover, women tend to experience higher rates of food insecurity, even when controlling for household income levels, reflecting deeper gender inequalities in the sector (Niranjan, 2024a, 2024b). The income gap between large and small farms has been growing, with large farms benefiting from economies of scale while small farms struggle to compete

(Niranjan, 2024a, 2024b). Over the past 15 years, the income disparity has doubled, putting immense financial pressure on smaller farms. Many small farms are closing due to minimal profit margins, and farmers are increasingly seeking alternative income sources to stay afloat. Both Ireland and Italy identify young farmers, women and small farms as the groups most affected by the transition to more sustainable and digitalised practices. In the period 2005-2020, the proportion of young farmers in the overall farming population declined in the EU. The trend differs among Member States. On average the share of female young farmers is especially low. (D3.1, pp. 91-92).

7.2.3 DGs identified in T3.2

In T3.2, 11 interviews with experts were carried out to better understand the twin transition and identify potentially disadvantaged groups in the context of the twin transition. The interview findings on the potentially disadvantaged groups identified are outlined below.

Energy

Potential disadvantaged groups detected:

- Low-skilled workers in extractive industries (coal, oil, and gas sectors) facing job losses.
- Workers in fossil-fuel-related infrastructure (boiler installers, gas station workers) losing employment due to the new 'energy industry'.
- Highly specialised workers in outdated technologies (e.g., certain solar panel models) who may struggle to adapt to newer innovations.
- People with low energy literacy who may find it difficult to navigate subsidies, new tariffs and smart energy management systems. More susceptible to manipulation and/or scams."

Transport

Potential disadvantaged groups detected:

- Unskilled workers from automotive factories ("blue-collar")
- Mechanics without proper knowledge to repair electric motors
- The whole value chain of European automotive factories (Europe has too much dependency on third countries to obtain batteries. Also, the European value chain is alive thanks to the fact that it is subsidised)
- Those users that do not have or can't have access to charging points. So, mainly, those who do not have a private garage spot. Especially relevant for those citizens that live in urban areas (less space, more competition for charging spots) and have to work outside the city (which can access only by car)
- Those users without enough capital for the initial acquisition of an electric vehicle
- Users without bureaucratic/grants knowledge, or with bureaucratic difficulties to access grants for EV acquisition
- The whole cobalt value chain, as producers are stepping away from it. We do not know if this is FITTER scope.

Building & Housing

- Groups of people that do not have the necessary resources to have an initial investment (for sustainable architecture, both rehabilitating and retrofitting).
- Grants and subsidies for that have multiple issues. They often require (1) a good bureaucratic knowledge from the beneficiary to do a subsidy request (2) a good knowledge of what subsidies can be requested (3) still enough money to make an initial investment, as they are often paid after the works,
- Older people are those who may have the most difficulties in using advanced technologies related to climate-adaptive housing, but they are also who are more likely to have capital for investment. Role of educational background.

Food & Agriculture

Potential disadvantaged groups detected:

- Small farmers who fail to digitalise and lack a business mindset in digitalisation.
- Older farmers who struggle to adopt new technologies and digital business models and/or are resistant to change.
- Consumers who are highly sensitive to food quality and price fluctuations.
- Manual crop workers, particularly seasonal and migrant workers in rural areas, who may be displaced due to automation and have not been reskilled.

7.2.4 DGs identified in ongoing sectoral case study work

Ongoing sectoral case study work is proceeding to identify DGs in the context of the twin transition. Case studies are developed starting from policy objectives that were identified as common to the different countries in each case study (in T3.1). Following the choice of policy objectives, policy mechanisms were identified (T3.2 interviews). Then, case study reports for each sector examined potential DGs, and are closely linked to the objectives and mechanisms studied (they would be different if another objective or mechanism was used to determine the scope of the analysis).

Energy

The policy objective identified for Energy is the transition to Renewable Energy Sources (RES). The mechanisms considered are those that support programmes for decentralised energy (at the community and household level).

Considering the overall objective of the use of decentralised renewable energy, and the specific issues derived from the Irish policy and supports, the following groups should be considered as target groups for FITTER – with both taken to account, those who are targeted by the support programmes, and those missing from the policy action:

- Individual households who can directly access the supports for micro-generation: who are target for the micro-generation, with further examination of which socio-economic groups can disproportionately benefit from the programme (e.g. home owners, particularly those with detached houses, and with higher income/assets) and those who would need further supports to engage with the action (lower-income groups, those living in multi-flat buildings).

Households who are asset and income rich, but lack digital literacy (e.g. older citizens) also as a possible target group to engage with.

- Individual households who are not directly a target of the supports: those who are excluded from the programmes, for example tenants in private owned properties, tenants in housing associations and tenants in social housing – also as possible participants in community-level programme.
- Housing associations, voluntary housing bodies and other small-scale (local and neighbourhood based) groups which are potentially participating in the community-level small-scale generation scheme (to assess enablers and barriers)
- Local SMEs and CSOs who are potentially participating in the small-scale generation along with homeowners or tenant associations, particularly in the disadvantaged communities where the financial capacity at the household level is limited. (National Case Study: Energy Sector Summary, p. 12).

Transport

The National Case Study: Mobility and Transport Sector summary uses the following table to list potential structurally disadvantaged groups related to the policy objective (transition to electric mobility) and mechanisms identified (supports for transition from combustion engines to electric mobility) (National Case Study: Transport Sector Summary, p. 16).

	Economic Affordability	Social Accessibility	Geographical Accessibility	Reliability and EHS
Spain	Low-income households, households in smaller towns and rural areas, individuals reliant on older, cheaper vehicles.	Individuals without access to private parking or charging, elderly individuals or those with mobility issues. Digital dimension Individuals with limited digital literacy or internet access, individuals who prefer or rely on cash payments for services like charging.	Residents of rural and remote areas, communities with limited public transport options.	Communities near battery plants (experiencing environmental and health concerns), Potential EV adopters concerned about the reliability of charging infrastructure.
Hungary	Low-income households, households in smaller towns and rural areas, individuals reliant on older, cheaper vehicles.	Individuals without access to private parking or charging, elderly individuals or those with mobility issues. Digital dimension Individuals with limited digital literacy or internet access,	Residents of rural and remote areas, communities with limited public transport options.	Communities near battery plants (experiencing environmental and health concerns) Potential EV adopters concerned about the reliability of charging infrastructure.

		individuals who prefer or rely on cash payments for services like charging.		
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Table 13. Potential structurally disadvantaged groups in relation to the transition to electric mobility

Building & Housing

The policy objective identified for Building & Housing is the development of the country’s housing market (in terms of the social dimension, economic dimension and climate dimension), and the chosen mechanisms are related to this objective (supports for decarbonising the building sector).

Operational Dimensions to be considered: (1) Affordability, (2) Social accessibility, (3) Geographical accessibility, (4) Reliability and EHS (see risk matrix proposed in methodology workshop slides: Methodology Step By Step).

- Affordability
 - The national emissions trading system for heating and transport (BEHG) will impose greater burdens on households with low incomes than on households with high incomes (German Federal Government, 2022). This is being countered with one relief measure: from an incomes point of view, lower-income groups will benefit more from the abolition of the renewable energy surcharge (EEG surcharge) than higher-income groups. Its early reduction to zero in 2022 and final abolition in 2023 have therefore relieved a great deal of the strain on domestic customers (NECP, 2024).
 - The burden of housing and heating energy costs can be linked relatively clearly to the employment status of households. Among low-income households, pensioners, students, trainees and the unemployed are much more common, i.e. people who are not fully employed per se. Age also plays a role here. Younger and older people are less likely to have a high income. The former are often still in training or studying, while the income of older people over 65 drops permanently when they retire. Only retired civil servants continue to have a high income level even in old age due to the special regulations. The group in the bottom thirty per cent by income also very often includes single-person households and single-parent households. Single women in particular often have a low income, while a much smaller number of single women can be found in the group with a medium or higher income. On the other hand, higher incomes are much more common in households with self-employed people, civil servants and retired people. (DENA, 2024)
 - The inequality of the housing cost burden between income groups is much more pronounced for renters than for homeowners. On average, renting households with low incomes spend a much higher proportion of their income on housing than those with high incomes. This does not apply as clearly to owner-occupiers. Although in both cases the burden is also an expression of personal preferences such as location, quality or furnishings, tenants have to take greater account of what is on offer, especially in tight housing markets, while the level of the credit burden of owner-

occupiers is more a matter of free choice. As a rule, the amount of a loan taken out is primarily based on disposable income and the desired lifestyle. (DENA, 2024)

- Social accessibility
 - there is the need to check who can apply for these grants, if for example you need to have the German nationality or if being registered is enough
 - Bureaucratic understanding and digital skills are needed to fill out the applications for the funding schemes, which is particularly difficult for people socialised outside of Germany, older age people, people with lower formal education. Digitalisation in renovated/newly constructed homes often need a high digital literacy as well to use the house in an efficient way.
 - People, who have not inherited from their families, often start off with way less financial investment capital and are therefore only able to afford a home later in life or are bound to stay in tenancy contracts and are therefore at higher risk for tenancy law breaches, gentrification and renovation processes. Oftentimes racialised people and people with migration biography are less likely to inherit money or estate from their parents/grandparents.
- Geographical accessibility
 - people in urban areas/most populated areas are the people who most often rent their apartments
 - in urban areas a higher share of the rented apartments is owned by mega-landlords, private companies, oftentimes renting out to structurally vulnerable cohorts
- Reliability and EHS
 - Low-income groups, migrants, and especially homeless people at great risks since they are more often living in unrenovated, inefficient, non-permanent housing and are overlooked at by almost all of the measures offered by the Federal government of Germany.
 - Older people are especially at risk in regards to health aspects, as they often live in houses that are very big, but do not have the financial means to renovate them, resulting in heating (and cooling) them inappropriately/not sufficiently, leading to health issues.
 - Younger people more often cannot afford to live in a renovated or new building facing EP, health issues impacted by mouldy apartments. Further, their mental health is oftentimes affected by short-term leases or being financially dependent on their families.” (National Case Study: Building & Housing Sector Summary, pp. 16-18).

Food & Agriculture

The policy objective identified for this sector was rural development and digitalisation. Key mechanisms identified relate to supports for rural digital connectivity and digital skills development, particularly those targeted at farmers. Potentially disadvantaged groups identified in this context include low income farmers, older farmers, and farmers in isolated regions (National Case Study: Food & Agriculture Sector Summary, pp. 13-14).